
PERSONA PORTRAIT · 17 SECTIONS · ~307 ATTRIBUTES · MCKINSEY 4S STUDY

The Deadline-Bearing Committed STEM/ML Masterer

*A decision-grade persona — every attribute paired with the **Core Question** that produces it and an honest **Confidence** probability — plus a 262-entity ad-targeting channel & location map.*

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Confidential. Produced by the ritsu-works AI-Native operating company. Companion to customer-journey.pdf.

Persona Portrait — The Deadline-Bearing Committed STEM/ML Masterer

The decision-grade persona for Ritsu's locked first-100 wedge (scorecard winner 7.24/10; `/think mckinsey run ritsu-segment-persona-journey-v1`). Companion to `customer-journey.md`. **Every attribute carries (a) the Core Question that produces the answer — the question a stranger would ask to elicit it — and (b) an honest Confidence = the probability the answer is correct.** At true-zero, most answers are `hypothesis / inferred`; `observed` traces to the `supabase-analytics` Door-2 signal (25 profiles · 756 sources · 656 sessions · 15 founder-only shares · 0 real paying).

The portrait (in one breath)

It is week 3 of cs231n (or MIT 6.S191, or Andrew Ng's ML course). **The Masterer** — a sophomore-to-grad STEM student or an early-career engineer switching into ML, 18–32, laptop-first, studies late and alone — is staring at a backprop derivation they've re-read three times, a graded midterm N weeks out, and the old whisper: *"maybe I'm just not a math-science person."* They are wrong, and the deepest hook in the whole product is telling them so: **the method was wrong (passive re-reading), not you.** They don't want an AI to *do* it — they want to be *sharper*; they respect rigor and instantly distrust gimmicks. They upload a dense PDF or a long lecture video (PDF 63% + YouTube 16% — **observed**), revisit it across many sessions to actually master it (one real user: 42 sources / 36 sessions — **observed**), and would refer a classmate the moment a result is legibly theirs. Their willingness to pay is **real but deadline-gated** — sharp the week before a graded checkpoint, soft between them — and it is the **one thing never yet observed** (0 real paying). Their shadow rival is **free**: Google NotebookLM now does doc→quiz/flashcards→grounded-explanation→share-link, and as of **April 2026** even basic mastery-tracking — so Ritsu wins *only* on the structured multi-week **PATH** + 17 activity types + concept-level mastery map + learning-science, never on "we also make quizzes."

- **One-line definition:** *A serious learner gripping a hard, multi-week, graded STEM/ML course who must TRULY master dense material before a real deadline — and will pay to actually learn, not just pass.*
- **The 2 load-bearing risks (carried throughout): (R1)** WTP is unproven at true-zero — the N=10 watch must test *"do they pay at the first hard limit?"*; **(R2)** free NotebookLM clones the core spine (and narrowed the gap further in April 2026) — wedge + message must lead with the PATH/ mastery/learning-science it lacks.

How to read this document

- **Core Question** column — the question (from the persona template) whose answer *is* this row. Read it as "to learn X about this customer, you would ask: ...".
- **Conf** column — **the probability the answer is correct**, with its evidence tier:
- **Obs · 80–92%** — grounded in `supabase-analytics` Door-2 behavioral data (small-N caps it below ~92%).
- **Inf · 55–72%** — reasoned from Kotler / 5-Star BRAVO / JTBD frameworks + market deep-research (triangulated, but not directly observed for *this* segment).
- **Hyp · 30–50%** — plausible and internally consistent, but untested; would move on contact with N=10.

- ★ before the row number = one of the **load-bearing (critical) attributes** — the ones that directly predict whether the first-100-paying bet works (mirrors the template's *** flag; ~110 of ~307 are critical).
- Anything touching **payment, organic referral, or "beating free NotebookLM"** is honestly low-confidence — that is the unproven core, not a hedge.

S1 · Segment Identity & Scope

(Kotler STP Ch2 + Ritsu wedge · 14 attrs · 4★)

#	Attribute	Core Question	Answer	Conf	Source
1	Segment name	<i>What single name captures who this is?</i>	The Deadline-Bearing Committed STEM/ML Masterer (ML/AI-course beachhead → broader STEM expansion)	Inf · 70%	locked segment + icp-summary
2	One-line definition	<i>Can you define them in one sentence a stranger understands?</i>	A serious learner gripping a hard multi-week graded STEM/ML course (cs231n, fast.ai, Andrew Ng) who needs to TRULY master dense material before a real deadline	Inf · 70%	locked def + Kotler STP
★3	Hypothesis status	<i>Is this segment a guess, an inference, or observed truth?</i>	Engagement OBSERVED (sustained revisit: one user 42 sources/36 sessions); PAYMENT untested (0 real paying, 2 founder test)	Obs · 88%	analytics live.sessions/sources
★4	Wedge vs expansion	<i>Is this the wedge or the expansion ladder?</i>	WEDGE = deadline-bearing committed masterer (durable graded-deadline WTP); EXPANSION = open-courseware self-learner (huge/reachable, soft WTP, top-of-funnel only)	Inf · 68%	locked PAYING-CORE vs REACH split
5	TAM / SAM / SOM	<i>How big is the reachable, winnable slice?</i>	TAM = global English STEM/MOOC learners (10s of M); SAM = active ML/STEM course communities (r/ML ~3.1M + r/GradSchool ~924K + r/learnmath ~203K + fast.ai/cs231n ≈ 4–5M reachable); SOM = first-100 paying	Inf · 62%	WebSearch: subreddit counts
★6	Priority rank	<i>Is this segment #1 to win, or later?</i>	#1 and ONLY first-100 target — concentrate 100% of GTM here; all other personas/VN = deferred expansion ladder	Inf · 72%	icp-summary concentration + north-star
7	Targeting strategy	<i>How broadly or narrowly do we target?</i>	Concentrated (single-segment); narrow to win passionate referring advocates over a one-shot crammer who churns	Inf · 70%	Kotler concentrated targeting
8	B2C / B2B / B2B2C	<i>Who is the buyer vs the user vs the payer?</i>	Pure B2C — buyer = user = payer (the individual learner); no institutional gatekeeper at this wedge (B2B educator path = S17 expansion only)	Inf · 72%	product.md §8 + self-serve
9	Intermarket reach	<i>Does this persona transfer US↔UK↔VN unchanged?</i>	US-LED intermarket; identical "master this course before the deadline" job spans US↔UK↔CA↔AU↔global English; VN = secondary served-market	Inf · 64%	icp-summary intermarket + §6.6
10	Persona mapping (6 marketing personas)	<i>Which of Ritsu's 6 marketing personas does this map to?</i>	Primary = Students (graded STEM); secondary = Self-Learners + Engineers (framework-from-docs); Researchers sit on the expansion edge	Inf · 66%	product.md §8 (6 personas)
★11	Anti-persona check	<i>Does this overlap an anti-persona we should NOT chase now?</i>	YES, and we exclude them: the one-shot exam-crammer (churns post-exam), the casual browser (no graded deadline), the reference-looker, and the edutainment/streak-seeker — same demographics, opposite payment behavior	Inf · 68%	locked wedge; avatar-split

#	Attribute	Core Question	Answer	Conf	Source
12	Positioning-belief fit	<i>Does it fit the "wrong method, not weak ability" belief?</i>	1:1 — the reframe ("you're not bad at this; passive re-reading is") is the segment's deepest hook; the belief defines the segment	Inf · 66%	positioning.md belief
13	Referral-speed fit	<i>Does the social context (study cohort) drive fast referral?</i>	High latent: a shared graded deadline gives built-in cohort recipients (course Discord, study group) — but stranger-referral is UNPROVEN (15 shares, all founder)	Hyp · 45%	analytics shares; cohort JTBD
★14	Strategic rationale	<i>Why this segment, why now?</i>	Only MASDA-eligible candidate: fuses sustained-mastery retention (anti-churn) with a graded-deadline WTP trigger, 1:1 product fit, pre-assembled audience; leans on the PATH + 17 activities + mastery-tracking free NotebookLM lacks	Inf · 70%	locked WHY-IT-WON

S2 · Geographic

(Kotler Ch7 geographic · 12 attrs · 1★)

#	Attribute	Core Question	Answer	Conf	Source
1	Country / market	<i>Which country are the first buyers in?</i>	US-led intermarket; paying core in the US, reach layer across global English (UK, CA, AU, India, SG, NG)	Inf · 65%	icp-summary; Coursera 2025 (US 27.7M, India 27M)
2	Region / state	<i>Which regions concentrate the segment?</i>	US coastal + research-university clusters: CA (Stanford/Berkeley), MA (MIT), NY, WA, TX	Hyp · 45%	Kotler Ch7; beachhead-course inference
3	City tier / metro	<i>Do they cluster in specific city types?</i>	Tier-1 university metros (SF Bay, Boston, Seattle, NYC, Austin) + global uni hubs (London, Toronto, Bangalore)	Hyp · 42%	deep-research MOOC hubs
4	Urban vs rural	<i>Urban, suburban, or rural?</i>	Overwhelmingly urban/campus-adjacent; broadband-dependent dense-info workflow rules out rural-constrained	Inf · 66%	analytics (PDF 63% +YT 16%)
5	Language(s)	<i>What language must the product/marketing be in?</i>	English-first (product belief + creators are English: Sung/Abdaal/Frank); consumed in English even by ESL	Obs · 85%	analytics (EN material/creators)
6	Time zone	<i>When are they online and studying?</i>	US ET-PT core; reach adds GMT/IST → 24h support + async, no single peak	Inf · 60%	icp-summary intermarket
7	Climate / seasonality	<i>What calendar drives their need?</i>	Demand pulses on the US academic calendar: midterms Oct+Mar, finals Dec+May; cohort/bootcamp deadlines add spikes	Inf · 64%	Kotler Ch7; deadline-bearing JTBD
8	Market maturity	<i>Is the category new or saturated there?</i>	Mature, saturated EdTech — heavy incumbent stack (Anki/Quizlet/ChatGPT/NotebookLM); enter on differentiation, not category creation	Inf · 68%	NotebookLM scan
9	Regulatory zone	<i>Which legal regimes govern serving them?</i>	CCPA + US state-privacy for core; GDPR for UK/EU reach; COPPA edge (some <18 HS-AP) — adult-default + age-gate	Inf · 66%	product.md §12
★10	Payment geography	<i>Can they actually pay you, and how?</i>	Card-payable USD via LemonSqueezy MoR (US tax + intl); \$29 Plus frictionless US, higher relative barrier India/NG	Obs · 84%	analytics live.payments (LemonSqueezy)

#	Attribute	Core Question	Answer	Conf	Source
11	Connectivity	<i>Will bandwidth/devices constrain the experience?</i>	High-bandwidth assumed: 16% YouTube + dense PDFs need stable broadband; not low-data/mobile-only	Obs · 82%	analytics live.sources
12	Localization need	<i>How much must we localize vs reuse?</i>	Near-zero for paying core (English UX); reach-layer ESL self-serve in English — defer 9 locales until post-PMF	Inf · 64%	icp-summary English-first

S3 · Demographic

(Kotler Ch7 + Ch3 beyond birth year · 24 attrs · 2★)

#	Attribute	Core Question	Answer	Conf	Source
1	Age band	<i>What age range are they in?</i>	18–32 core; bimodal — 19–24 (university STEM/cohort) + 24–32 (career-switch ML self-learner); thin <18/>35 tail	Inf · 64%	course-tier composition + Kotler Ch7
2	Generation cohort	<i>Which generation, and does it actually matter here?</i>	Gen Z dominant (1997–2006 undergrad core) shading into younger Millennial (1990–96) upskiller; matters for tone, not for the JTBD	Inf · 60%	Kotler cohorts
3	Life stage	<i>What life stage shapes the need?</i>	Mid-degree student OR early-career professional retraining into ML/AI; identity-forming "becoming technical" phase	Inf · 66%	icp-summary + JTBD
4	Gender mix	<i>Does gender change the job or the message?</i>	Skews male (~60–70%, mirrors STEM/CS) but the mastery-anxiety belief is gender-neutral; do NOT message male-only	Hyp · 48%	deep-research STEM skew; icp-summary
5	Education level	<i>What is their education level?</i>	Bachelor's in-progress (paying core) to early-grad; high baseline literacy, comfortable with dense PDFs/papers	Obs · 80%	analytics (PDF 63% + paper material)
★6	Field / discipline	<i>What are they studying or working in?</i>	ML/AI beachhead (DL, CV, stats); expansion = calc/orgo/bio/physics + pre-professional STEM	Inf · 70%	locked beachhead; code+slides mix
★7	Academic load	<i>How heavy is their memorization/ mastery load?</i>	Very high MASTERY load (derivations, proofs, code, multi-week chains) — not memorization; "must truly get it, graded"	Obs · 84%	analytics (42 src/36 sess revisit)
8	Institution type	<i>Where does the learning happen?</i>	R1/competitive 4-yr universities + structured bootcamps/cohort courses (fast.ai, 6.S191) + grad programs; not casual MOOC drift	Inf · 64%	beachhead list; deadline-bearing core
9	Student year	<i>Freshman cramming or thesis-stage?</i>	Sophomore–senior (declared major, hard courses) for core; grad-quals takers; bootcamp cohort members	Inf · 58%	cs231n upper-div/ grad
10	Occupation	<i>What do they do for work?</i>	Full-time student OR SWE/analyst/researcher upskilling into ML; output is graded or job-gating	Inf · 64%	segment + sustained-mastery pattern
11	Employment status	<i>Are they earning, and how stably?</i>	Paying core = enrolled student + employed early-career switcher; both have a hard external deadline	Hyp · 50%	dual-core; deadline-WTP
12	Income band	<i>How much can they spend monthly?</i>	Tight: student \$0–300/mo discretionary; switcher \$500–1500/mo — both price-anxious, \$29 Plus must clear a felt-pain bar	Inf · 58%	icp-summary US WTP anchor

#	Attribute	Core Question	Answer	Conf	Source
13	Who funds them	<i>Whose money pays for tools?</i>	Self-funded (realistic default); some employer L&D for switchers; rarely parents — the decision is personal	Hyp · 48%	icp-summary price-sensitive
14	Household / family	<i>What home context affects study time?</i>	Solo focused study — dorm desk, shared apartment, or WFH corner; headphones-on deep-work, often late-night	Hyp · 44%	study-creator deep-work culture
15	Socioeconomic tier	<i>Are they price-sensitive by circumstance?</i>	Middle / aspirational-middle; education-as-mobility; will pay for a tool that protects a graded outcome	Inf · 60%	icp-summary mastery-over-grades
16	Device ownership	<i>What hardware do they own + actually use?</i>	Laptop-primary (PDF + code + long-video) + smartphone for review; <60s upload→path fits desk-first	Obs · 86%	analytics (PDF+YouTube+code)
17	Tech proficiency	<i>How easily do they adopt a new tool?</i>	High — runs Colab/Anki/ChatGPT/Notion; early-adopter; judges Ritsu against NotebookLM on day one	Obs+Inf · 78%	analytics (code uploads); NotebookLM scan
18	Accessibility needs	<i>Any accessibility requirement to design for?</i>	Mostly standard; latent demand for dyslexia-friendly reformatting + audio/visual variety (17 types serve this)	Hyp · 40%	product 17-activity fit
19	Ethnicity sensitivity	<i>Any cultural nuance the message must respect?</i>	Ethnically diverse (heavy international + Asian STEM); keep imagery inclusive, never persona-as-default-white	Hyp · 46%	deep-research STEM/intl enrollment
20	Nationality / residency	<i>Any residency constraint on access or payment?</i>	US-led intermarket: US lead + global English (India, EU, SEA, LatAm) reachable on Reddit/YouTube/Discord	Inf · 62%	icp-summary US-led
21	Marital / relationship	<i>Does relationship status change availability?</i>	Predominantly single / no dependents — time + money flow to self-investment; learning a top priority	Hyp · 46%	age band 18–32
22	Career stage	<i>Where are they on the career arc?</i>	Pre-career (building credential) OR early-career pivot (0–5 yrs) climbing into ML/AI; high upside-motivation	Inf · 62%	dual-core; "sharper not lazier"
★23	Self-view of field difficulty	<i>Do they secretly think their field is "too hard for me"?</i>	YES — the fragile "I'm not a math-science person" belief is the deepest psychographic hook; one bad exam reactivates it	Hyp · 50%	icp-summary reframe (VI-added)
24	Incumbent-tool proficiency	<i>Are they an Anki/Quizlet power-user, or a frustrated newbie?</i>	Mixed: ML self-learners are often Anki-literate; many STEM students are frustrated manual-card-makers (the grunt-work pain Ritsu removes)	Inf · 56%	analytics; icp stack (VI-added)

S4 · Psychographic (behavior drivers — the engine of the segment)

(5-Star BRAVO psychographics > demographics + Kotler Ch7 + Ritsu "wrong-method" belief · 29 attrs · 11★)

#	Attribute	Core Question	Answer	Conf	Source
★1	Core value	<i>What do they fundamentally value here?</i>	Mastery — truly <i>understanding</i> hard material, not memorizing for a grade; "I want to actually get it."	Inf · 70%	icp-summary + Kotler
★2	Core belief	<i>What do they believe about themselves and this problem?</i>	Active learning beats passive; doing problems/explaining-back creates mastery, re-reading wastes the hours before a deadline.	Inf · 68%	product.md §4 + BRAVO

#	Attribute	Core Question	Answer	Conf	Source
★3	The reframe	<i>What false belief can you flip into relief?</i>	"I'm not a math/science person" → relief: the METHOD was wrong (passive re-reading), not you; with active recall you CAN master ML.	Hyp · 48%	icp-summary psychographic core
4	Self-image	<i>Who do they think they are?</i>	A serious, capable-but-struggling learner who out-works rather than out-talents; "I grind hard, I just need the right system."	Inf · 62%	icp-summary + scorecard
5	Aspirations	<i>What do they aspire to be?</i>	Pass the graded course/quals, land the ML/SWE role or grad slot, become someone who genuinely understands deep nets — not a faker.	Inf · 64%	beachhead courses
★6	Deepest fear	<i>What are they most afraid of?</i>	Putting in the weeks, hitting the midterm/final, and still failing — proving the "not smart enough" story true.	Hyp · 50%	JTBD + deadline trigger
7	Secondary fears	<i>What lesser fears nag at them?</i>	Wasted study time, falling behind a cohort, an AI feeding them a wrong fact that tanks an exam answer.	Hyp · 46%	NotebookLM scan + SERVQUAL
★8	Motivation type	<i>Are they driven by mastery or by a grade/deadline?</i>	DUAL: intrinsic mastery-drive (durable LOVE) GATED by a recurring graded deadline (the WTP trigger) — not pure cram.	Inf · 66%	locked wedge
9	Growth mindset	<i>Do they think they can get better?</i>	High but fragile — believes effort compounds, yet one bad exam reactivates the fixed "not-a-math-person" narrative.	Hyp · 48%	icp-summary reframe
10	Risk tolerance (new tool)	<i>How much risk will they take on a new tool?</i>	Moderate — will trial mid-course if onboarding is <60s and doesn't risk their limited deadline hours.	Inf · 60%	magic-moment + deadline pressure
11	Novelty seeking	<i>Do they chase new tools or distrust them?</i>	Low-to-moderate for tools (rigor > shiny); high for <i>insight</i> — loves an explanation/analogy that makes a hard concept click.	Hyp · 46%	icp + mastery engine
12	Status sensitivity	<i>Does looking capable to peers matter?</i>	Quiet/competence-based — respected for understanding, indifferent to gamified badges; "studytwit streak" optional flex.	Hyp · 44%	#studytwit + mastery- over-grades
13	Locus of control	<i>Do they feel in control of results?</i>	Strongly internal — outcomes hinge on their own study system, which is exactly why a method-fixing tool resonates.	Inf · 64%	BRAVO + icp
★14	Conscientiousness	<i>How organized and disciplined are they?</i>	High — uploads dense multi-session material and revisits (42 sources/36 sessions; 4 sources/10 sessions = deliberate mastery loops).	Obs · 86%	analytics live.sources/ sessions
15	Procrastination	<i>Do they act early or only under deadline?</i>	Present but deadline-bounded — the graded midterm/final is the forcing function that converts intent into sustained sessions.	Inf · 60%	deadline trigger + 656 sessions
16	Perfectionism	<i>Does perfectionism help or paralyze them?</i>	Moderate-high on understanding, pragmatic on polish — depth over completeness under time pressure.	Hyp · 44%	mastery-over-grades
★17	Attitude to AI	<i>Do they trust AI, or fear it "cheats/hallucinates"?</i>	Conditional trust — adopts AI for leverage but viscerally fears hallucinated quizzes; demands source-grounded output.	Inf · 66%	icp + NotebookLM scan + SERVQUAL
★18	Attitude to effort	<i>Do they want a shortcut or a sharpening?</i>	Wants AI to make them SHARPER, not do it for them; rejects "AI does it for you" — effort IS the point.	Inf · 68%	icp + "AI can't learn for you"
19	Identity tribe	<i>Which tribe do they identify with?</i>	Open-courseware ML self-learners + university STEM grinders — r/learnmachinelearning, fast.ai forums, cs231n cohorts, #studytwit.	Inf · 62%	channels + reach layer

#	Attribute	Core Question	Answer	Conf	Source
20	Lifestyle	<i>What does a normal day look like?</i>	Time-poor, deadline-stacked: full course load / bootcamp weeks / job-plus-study; long focused multi-source sessions.	Inf · 60%	analytics session pattern
★21	Values hierarchy	<i>Which values win when they conflict?</i>	Mastery > accuracy/trust > time-saved > cost > status/gamification — pays for depth, not dopamine.	Inf · 66%	icp + SERVQUAL + pricing
★22	Skeptic triggers	<i>What instantly makes them roll their eyes?</i>	A hallucinated/wrong quiz; generic non-grounded answers; "AI shortcut" framing; flashy gimmick over rigor; paywall before first value.	Inf · 66%	NotebookLM scan + icp
23	Emotional before-state	<i>How do they feel right now, unsolved?</i>	Anxious + overwhelmed — dense material, ticking deadline, secret doubt they're "not smart enough."	Hyp · 50%	JTBD + deepest fear
24	Emotional after-state	<i>How do they want to feel once solved?</i>	Relief + earned confidence — "it finally clicked, I can actually do this," the method (not their brain) was fixable.	Hyp · 48%	reframe + magic-moment
★25	Self-talk / language	<i>What words do they actually say to themselves?</i>	"I keep re-reading and nothing sticks." / "I need to actually understand this, not just pass." / "Am I just not a math person?"	Hyp · 50%	icp verbatim + JTBD
★26	Avatar split risk	<i>Could identical demographics hide opposite avatars?</i>	HIGH: paying committed-masterer (durable WTP) vs soft-WTP open self-learner (free reach) — same psychographic, opposite payment behavior; courting reach can dilute the paying core.	Inf · 66%	locked reach-vs-core + R1
★27	Attitude to "AI cheating"	<i>Do they use AI to AVOID the work, or to get sharper at it?</i>	To get sharper — they reject "AI does it for me"; using AI to skip understanding would defeat their own goal. The anti-cheat stance IS the buy reason (Cohere belief: "AI can't learn for you").	Inf · 66%	icp belief (VI-added)
28	Learning-science affinity	<i>Have they already met active-recall / spaced-repetition?</i>	Moderate-to-high: many already run Anki / know spaced repetition; naming the science (Feynman, retrieval practice) is a <i>trust signal</i> , not a novelty — the preemptive moat.	Inf · 58%	positioning learning-science (VI-added)
29	Desired identity shift	<i>What identity do they crave — from "X" to what?</i>	From "I'm not a math-science person / a faker" → "someone who genuinely understands this" — a competence identity, earned not faked.	Hyp · 46%	reframe + aspirations (VI-added)

S5 · Jobs-to-be-Done

(JTBD 3-layer + Ritsu icp · 21 attrs · 9★)

#	Attribute	Core Question	Answer	Conf	Source
★1	Functional job	<i>What practical job do they need done?</i>	Convert a dense multi-week STEM/ML course (PDF+lectures) into recallable, exam-ready mastery before graded midterms/finals	Obs · 84%	analytics (PDF 63% +YT 16%)
★2	Emotional job	<i>What feeling are they hiring for?</i>	Replace "I'm not a math-science person" dread with earned confidence — feel genuinely capable, not just busy re-reading	Inf · 64%	icp-summary (false-belief)
3	Social job	<i>How do they want others to see them?</i>	Be seen by peers/professor/cohort as someone who actually <i>gets it</i> — can explain it back, not just pass	Hyp · 46%	JTBD; #studytw norms

#	Attribute	Core Question	Answer	Conf	Source
★4	Struggling moment	<i>When exactly does the pain become urgent?</i>	10 days pre-final staring at a 200-page reader + 12 lecture hours, re-reading for the 3rd time and still not retaining	Obs · 80%	analytics (42 src/36 sess revisit)
5	Desired progress	<i>What does progress look like to them?</i>	From passive "I've read it" to active "I can solve it cold under time pressure" — durable cross-topic mastery, not cram-and-dump	Inf · 64%	icp-summary; mastery engine
6	Current workaround	<i>What hacky solution do they use now?</i>	DIY frankenstack: ChatGPT paste + manual Anki + Quizlet + YouTube + re-reading PDFs — fragmented, no path, no tracking	Inf · 66%	product.md (\$25–100 stack)
★7	Hire trigger	<i>What pushes them to try something new?</i>	A real graded deadline lands (midterm/final/qual date) on material too dense+long to brute-force by re-reading	Hyp · 50%	locked wedge (deadline = WTP trigger)
★8	Fire trigger	<i>What makes them quit a tool?</i>	Course ends / deadline passes, OR a hallucinated quiz breaks trust in accuracy (#1 SERVQUAL bar)	Inf · 62%	R1 + product.md (accuracy #1)
9	Success criteria	<i>How will they know it worked?</i>	Aced the graded assessment AND can still apply the concepts weeks later — mastery that survived the exam	Hyp · 48%	icp-summary (mastery-not-cram)
★10	Force: Push	<i>What pain pushes them away from today?</i>	Re-reading doesn't stick, time is finite, stakes are real — the passive method is visibly failing as the deadline nears	Obs · 80%	analytics (heavy revisit)
★11	Force: Pull	<i>What pulls them toward the new way?</i>	Upload → 30s structured multi-week path + 17 activities + mastery-tracking — the <i>whole</i> loop NotebookLM lacks	Inf · 64%	product.md; NotebookLM scan
12	Force: Inertia	<i>What habit keeps them stuck?</i>	"ChatGPT + Anki already works well enough" — the free frankenstack is owned, familiar, sunk-cost'd	Hyp · 50%	JTBD; self-learner defaults
13	Force: Anxiety	<i>What fear holds them back from switching?</i>	"Is the AI accurate, or will it teach me wrong things before my exam?" — distrust of AI gimmicks at high stakes	Inf · 64%	icp-summary; R2
14	Job frequency	<i>How often does this need arise?</i>	Recurring during a term: weekly study cycles + 2–4 graded deadline spikes/semester	Inf · 64%	locked wedge (recurring deadlines)
15	Job context	<i>In what setting does the job occur?</i>	Solo, late-night, laptop, deep-focus blocks; sourced from r/learnmachinelearning (~400K+), fast.ai forums, cs231n cohorts	Inf · 60%	analytics (656 sessions); deep-research
16	Job importance	<i>How much does getting this right matter?</i>	Top-3 life priority in-term — grade/credential/career gate; mastery is non-optional	Hyp · 50%	locked wedge (real stakes)
17	Little vs big hire	<i>What is the smallest first "yes"?</i>	Little: upload one lecture PDF, get a quiz in 60s (magic moment). Big: run a full course as a tracked multi-week path	Obs · 80%	analytics (32 projects, 3329 units)
★18	Non-consumption	<i>Are they ignoring the problem entirely today?</i>	Today they default to free re-read+ChatGPT and ignore paid tutors — the watch must prove they PAY at the first hard limit	Hyp · 42%	R1 (WTP unproven)
19	Success metric they track	<i>What number do they personally watch?</i>	The graded score (pass/A) externally; internally "can I solve an unseen problem cold?" + practice-quiz accuracy + felt-confidence	Hyp · 44%	JTBD (VI-added)
★20	Multi-source job	<i>Do they fuse multiple sources into one mastery project?</i>	YES — a hard course = textbook + slides + lecture video + problem sets fused into ONE tracked path (32 projects / 3,329 units observed)	Obs · 80%	analytics multi-source projects (VI-added)

#	Attribute	Core Question	Answer	Conf	Source
21	Passive habit to replace	<i>Which passive ritual must Ritsu replace?</i>	Re-reading + highlighting + passive lecture re-watching — the rituals that feel productive but don't stick	Inf · 64%	icp passive-method (VI-added)

S6 · Behavioral

(Kotler Ch7 occasions/usage/loyalty · 24 attrs · 9★)

#	Attribute	Core Question	Answer	Conf	Source
1	Usage occasion	<i>On what occasion do they reach for this?</i>	Triggered by a graded checkpoint: a problem set due, a cs231n midterm in 2 weeks, a fast.ai module, or quals prep	Inf · 64%	icp-summary + Kotler
★2	Benefit sought	<i>Which benefit do they most want?</i>	Durable mastery that survives the exam: explain-it-back, solve unseen problems, connect concepts — not passive recall	Inf · 66%	icp-summary (mastery-over-grades)
3	Secondary benefits	<i>What additional benefits sweeten it?</i>	Time-compression on material creation (auto-quizzes from a 60-pg PDF), confidence, one stack replacing Anki+Quizlet+ChatGPT	Inf · 64%	product.md §6 + substitution stack
4	User status	<i>What is their relationship to the category today?</i>	First-time / early-trial at true-zero: 25 profiles, 24 non-founder, ZERO real paying — engaged but unconverted	Obs · 88%	analytics (25 profiles, 0 paid)
★5	Usage rate	<i>How intensely would they use it?</i>	Bimodal: a heavy-mastery tail (36 sessions/42 sources; 10 sessions/4 sources) above a light-trial body; mean ~26 sessions/user	Obs · 84%	analytics (656 sessions/25 users)
★6	Loyalty status	<i>How loyal are they to current tools?</i>	Unproven, shifting: engagement real but no organic referral, no payment; loyalty rests on recurring-deadline cadence, undemonstrated	Hyp · 42%	analytics (0 organic share) + R1
★7	Buyer-readiness stage	<i>How ready are they to buy right now?</i>	Aware + engaged (Act) but pre-Conversion: nobody has hit-and-paid-through a hard limit; the N=10 watch must move Act→Pay	Inf · 58%	Kotler 5A + R1
8	Decision frequency	<i>How often do they reconsider their tools?</i>	Subscribe-decision term-gated: re-evaluated each course/semester (~2-4×/yr) at the next graded deadline	Hyp · 50%	icp-summary (deadline trigger)
9	Seasonality of use	<i>What calendar spikes their usage?</i>	Sharp spikes at midterm weeks (Oct/Mar), finals (Dec/May), quals windows, cohort-launch dates — trough over breaks	Inf · 64%	Kotler Ch7 + calendar
10	Session length	<i>How long is one sitting with the tool?</i>	Long deep blocks: 30-90 min/session implied by 656 sessions absorbing 756 sources + 3329 generated units	Inf · 60%	analytics (units/session ratio)
★11	Session cadence	<i>How often do they come back?</i>	Multi-session-per-source revisit (42 sources across 36 sessions) — returns to the same material to drill: the mastery tell	Obs · 84%	analytics (revisit ratio)
12	Time of day	<i>What hours do they study/use?</i>	Evening/late-night blocks + pre-class cramming windows; weekend long-form review — student maker-schedule	Hyp · 46%	JTBD (student rhythm)
13	Multi-tasking	<i>Do they use it solo or in a stack?</i>	Solo deep-focus, single-source-at-a-time; Ritsu as the primary surface replacing a tab-stack, not run alongside	Inf · 58%	product.md (replaces stack)

#	Attribute	Core Question	Answer	Conf	Source
★14	Trial behavior	<i>How do they trial a new tool?</i>	Uploads a real high-stakes doc first session (PDF 63% / YouTube 16%) to test the <60s magic moment — dense, not a toy file	Obs · 84%	analytics (live.sources mix)
15	Feature-adoption pattern	<i>Which feature hooks them first?</i>	Quiz + flashcard hook first (active recall); mindmap/Knowledge-Map + heatmap adopted later as mastery-tracking matters	Inf · 60%	product.md §6.2 + JTBD
★16	Abandonment triggers	<i>What makes them drop off?</i>	A hallucinated/wrong quiz (kills #1 accuracy bar) or hitting Free's 40-page/5-session wall mid-course with NotebookLM-free as the off-ramp	Inf · 62%	product.md §6.7 + R2
17	Reactivation triggers	<i>What would pull them back?</i>	The next graded deadline (new midterm/course) + a 7-day-silence nudge tied to an unfinished learning path	Hyp · 50%	icp-summary + SOP-CUSTOMER-014
★18	Referral behavior	<i>Do they tell friends, and how?</i>	Latent, unproven: share mechanism works (15 shares) but ALL founder — zero organic stranger referral observed	Obs · 88%	analytics (15 shares, all founder)
19	Review behavior	<i>Will they leave a public review?</i>	Rigor-respecting, skeptical: validates accuracy before trusting; posts in r/learnmachinelearning / #studytwit only after the tool proves it isn't a gimmick	Hyp · 46%	icp-summary
★20	Content-consumption	<i>How do they consume content?</i>	Dense multi-session inputs: PDF 63% (textbooks/papers) + long-form YouTube 16% (lectures) + URL 9% + slides/code	Obs · 86%	analytics (ex-founder mix)
★21	Sharing behavior	<i>Do they share results with peers?</i>	Generates share-links (15 total) — study-group/cohort sharing is the intended viral loop, strangers-referral UNPROVEN beyond founder	Obs · 86%	analytics (shares=15, founder-only)
22	Switching frequency	<i>Are they tool-loyal or tool-promiscuous?</i>	Low intra-course (sunk-cost of generated units + cross-session memory), but high pre-commitment shopping vs NotebookLM/Anki	Hyp · 50%	R2 + switching-cost inference
★23	First-session behavior	<i>Do they hit the magic moment in session #1?</i>	They upload a real dense doc immediately and the first-quiz aha is make-or-break; a weak first generation on hard math = instant churn	Obs · 82%	analytics (dense first uploads) (VI-added)
★24	7-day re-upload proxy	<i>Do they upload a 2nd source within 7 days?</i>	THE strongest observable retention signal — a 2nd upload within 7 days = mastery intent (top users 42src/36sess); the N=10 watch instruments it	Obs · 84%	analytics; SOP-PRODUCT-002 (VI-added)

S7 · Customer Journey (5A summary)

(Kotler Ch14 Aware→Appeal→Ask→Act→Advocacy · 16 attr · 6★ — full journey in [customer-journey.md](#) / [customer-journey.pdf](#))

#	Attribute	Core Question	Answer	Conf	Source
1	Awareness source	<i>How do they first become aware?</i>	Reddit (r/learnmachinelearning, r/MachineLearning ~3.1M; r/cs231n, r/MLQuestions) + a study-science YouTuber clip (Sung/Abdaal)	Inf · 62%	deep-research; icp-summary channels
2	Appeal trigger	<i>What makes them go "oh, I like this"?</i>	"Upload your cs231n PDF/lecture → a structured multi-week path + active recall in 30s," reframing the FALSE "I'm not a math person"	Inf · 62%	icp-summary; Kotler Appeal
★3	Ask behavior	<i>How much do they investigate before buying?</i>	Heavy comparison: Reddit + YouTube demos + side-by-side vs NotebookLM/Anki/ChatGPT; hunts "does it hallucinate quizzes?"	Inf · 60%	JTBD; NotebookLM scan; SERVQUAL

#	Attribute	Core Question	Answer	Conf	Source
★4	Act trigger	<i>What single thing makes them act?</i>	Sign-up: a real midterm/final/qualifier deadline. Pay: hitting the Free 600-credit/40-page wall mid-dense-PDF, deadline live	Hyp · 44%	locked WTP trigger; R1
★5	Advocacy behavior	<i>What makes them recommend it onward?</i>	Drops a share-link in their cohort Discord/group chat; mechanism works but ZERO organic stranger shares yet	Obs · 86%	analytics (15 shares all founder)
6	Journey duration	<i>How long is their path to buy?</i>	Long, deadline-paced: 3–14 days awareness→sign-up; convert-to-pay aligns to the multi-week course arc	Inf · 58%	analytics sustained pattern
7	Touchpoint count	<i>How many touches before they convert?</i>	6–9: Reddit/YT clip → landing → magic-moment demo → 1–3 upload sessions → free-limit wall → pricing → pay	Hyp · 48%	analytics; 5A funnel
8	Pre-purchase content	<i>What content do they consume before paying?</i>	"X vs Ritsu" thread; YT cs231n walkthrough; landing 30s-path demo; accuracy proof; pricing explainer; activity gallery; creator endorsement; free trial; a peer's share-link	Inf · 58%	icp-summary channels; Kotler content audit
★9	Drop-off points	<i>Where do they typically drop off?</i>	After magic-moment but before 2nd upload (no live deadline → no urgency); at pricing (\$29 vs free NotebookLM); soft-WTP reach-layer churns here	Hyp · 46%	analytics; R1; NotebookLM-free
10	Friction points	<i>What friction do they hit?</i>	First-upload quality on dense math PDFs (LaTeX/figures); credit-meter anxiety mid-PDF; "is this just ChatGPT?"; no social proof at true-zero	Inf · 60%	analytics PDF 63%; SERVQUAL
★11	Micro-conversions	<i>What small commitments precede paying?</i>	first upload → first path → first quiz → first revisit (session 2 = mastery) → first concept-map view → first share-link	Obs · 82%	analytics (656 sessions, 3329 units, 15 shares)
★12	Moment of truth	<i>What single moment decides it?</i>	First graded quiz on their OWN cs231n material returns accurate, source-cited questions that find a real gap	Inf · 62%	SERVQUAL accuracy #1
★13	Activation event (<60s)	<i>What is the magic moment for them?</i>	Upload dense lecture PDF → ~30s a structured path + first auto-generated quiz: "it built my whole study plan from my file"	Obs · 84%	analytics 30s path / 3329 units
14	Post-purchase emotion	<i>How do they feel after they pay?</i>	Relieved + capable ("I can actually master this before the deadline") — fragile if NotebookLM offers same free	Hyp · 46%	icp-summary; R2 (NotebookLM Apr-2026)
15	First-touch channel	<i>Which channel is the very first touch (for attribution)?</i>	A Reddit thread or a study/ML YouTube clip is usually first; a peer's Discord share-link is the warmest. Instrument first-touch for channel attribution (SOP-GTM-009)	Inf · 58%	channels; SOP-GTM-009 (VI-added)
16	Onboarding need	<i>Do they need a hand-onboard (Collison) or self-serve?</i>	First ~30 paying = founder hand-onboards (Collison install, SOP-CUSTOMER-006); the <60s magic moment is self-serve enough to automate scale onboarding from N=31	Inf · 60%	SOP-CUSTOMER-006 (VI-added)

S8 · BRAVO Positioning Fit

(5-Star Module 01 — Brand/Reason/Audience/Voice/Offering · 14 attrs · 7★)

#	Attribute	Core Question	Answer	Conf	Source
1	Brand fit	<i>Does the brand feel made for them?</i>	Strong: "active mastery, AI-leveraged" maps 1:1 to a deadline-bearing masterer who wants to be sharper, not have AI do the work	Inf · 66%	icp-summary; 5-Star Brand
2	Brand promise expected	<i>What promise must the brand make?</i>	"I'll actually understand this hard course before the midterm" — durable comprehension, not a crammed pass that evaporates	Hyp · 48%	JTBD; icp-summary
3	Reason-for-existing fit	<i>Does the company's why align with theirs?</i>	High: Ritsu exists to automate the painful part of active learning (making materials) so they spend time mastering — exactly this user's bottleneck	Inf · 66%	product.md §4
★4	Audience inclusion	<i>Why are they clearly our audience?</i>	Clearly ours: uploads dense PDFs + long ML videos, revisits heavily (42 sources/36 sessions)	Obs · 86%	analytics live.sources/sessions
5	Audience exclusion	<i>Are they in or explicitly out?</i>	NOT the one-shot crammer who churns post-exam, nor the casual browser with no graded deadline	Inf · 66%	locked wedge; cram-churn
6	Voice fit	<i>What tone earns their trust?</i>	Excellent: calm/rigorous/no-hype voice earns trust from a learner who distrusts gimmicks and respects rigor	Inf · 66%	icp-summary; brand_voice.md
★7	Offering fit	<i>Does what we sell match what they need?</i>	Strong: structured PATH + 17 activities + Knowledge Map serve a hard multi-week course; Plus \$29 vs \$25–100 stack	Inf · 64%	product.md; pricing
8	Positioning congruence	<i>Does the whole posture feel congruent?</i>	High: every BRAVO element points the same way — rigor, mastery, anti-gimmick — no internal contradiction	Inf · 64%	5-Star Module 01
★9	Differentiation perceived	<i>Why us over the alternative, in their words?</i>	"NotebookLM gives me quizzes/flashcards free — Ritsu gives the ordered multi-week PATH, 17 activity types, and concept-level mastery map it doesn't"	Inf · 64%	NotebookLM scan 2026
★10	Proof required	<i>What proof makes the claim credible?</i>	Accuracy receipts (no hallucinated quizzes on cs231n/orgo) + a visible mastery-progression heatmap they trust as real	Hyp · 48%	SERVQUAL §6.7
11	Before/after narrative	<i>What before→after story do they buy?</i>	Before: re-reading the same dense chapter, "not a math-science person." After: quizzed/explaining-back, the concept map fills in — "I get it"	Inf · 62%	icp-summary; product.md §7
12	Objection to position	<i>Why might they not believe our position?</i>	"Isn't this just ChatGPT/NotebookLM with a skin?" — answer with PATH + mastery-tracking + learning-science, not feature parity	Inf · 62%	NotebookLM scan; R2
★13	Message-market match	<i>What one line makes them lean in?</i>	"Turn your hardest course into a structured path that makes you master it before the deadline — not just cram it"	Hyp · 48%	locked wedge; positioning.md
14	Category frame	<i>What box do they put us in?</i>	"Personal AI mastery system for a hard course" — NOT a chatbot, flashcard app, or course platform; a structured-mastery loop	Inf · 64%	product.md §5; positioning.md

S9 · Channels & 'Locations'

(5-Star Traffic Lasso — Locations first · 18 attrs · 7★ — the deep per-platform ad map is S9-A below)

#	Attribute	Core Question	Answer	Conf	Source
★1	Primary platform	<i>Where do they spend most time online?</i>	Reddit — the default first-stop for course-specific Q&A, syllabus help, "how do I actually master this" threads	Inf · 64%	icp-summary; deep-research
2	Secondary platforms	<i>Where else do they hang out?</i>	YouTube (lecture rewatch — 16–17% of uploaded material), X/#studytwt, course Discords; PDF readers as the work surface	Obs · 80%	analytics (YouTube 16–17%)
★3	Communities	<i>Which communities are they in?</i>	r/learnmachinelearning, r/MachineLearning (~3.1M), r/cs231n, r/learnpython, r/GradSchool, fast.ai forums + per-cohort Discords (full list: S9-A)	Inf · 62%	WebSearch (r/ML 3.1M)
4	Influencers followed	<i>Who do they listen to?</i>	Study-science: Justin Sung, Ali Abdaal, Thomas Frank. ML: 3Blue1Brown, StatQuest, Andrej Karpathy, Andrew Ng, Jeremy Howard (full list: S9-A)	Inf · 62%	WebSearch; S9-A
5	Content formats	<i>What content format do they prefer?</i>	Long-form lecture video, dense PDF/slide decks, worked-problem walkthroughs, "how I studied for X" breakdowns, code notebooks	Obs · 82%	analytics (PDF 59–63% + YT + pptx + code)
★6	Search keywords	<i>What do they literally search for?</i>	"cs231n assignment help", "how to actually learn machine learning", "spaced repetition for STEM", "Andrew Ng notes quiz", "how to study for qualifying exams"	Inf · 60%	JTBD; deep-research
7	Search intent	<i>What are they really trying to find?</i>	Problem-solving + mastery-method intent ("I'm stuck / I keep forgetting / how do I retain this") — deadline-driven urgency	Inf · 60%	icp-summary; JTBD
★8	Discovery channel	<i>Where would they find us first?</i>	Organic search + community recommendation (Reddit thread, Discord drop, creator mention); share-links the planted seed once strangers convert	Hyp · 48%	Kotler 5A; analytics (15 founder shares)
9	Trusted info sources	<i>Whom do they trust for recommendations?</i>	The course's own materials + the professor, top Reddit/forum answers, named study-science creators; distrusts "AI does it for you" gimmicks	Inf · 62%	icp-summary
★10	Peer network	<i>Who do they ask before trying a tool?</i>	Coursemates / cohort-mates / study-group peers facing the SAME graded deadline — the natural recipients of a shared quiz/Knowledge Map	Hyp · 46%	locked wedge (deadline cohort)
11	Offline locations	<i>Any offline place that matters?</i>	University libraries, dorm/apartment study desks, lab sessions, coffee-shop cram sessions, bootcamp classrooms — solo deep-work	Hyp · 42%	persona inference
12	Ad-channel match	<i>Which paid channel matches their location?</i>	Reddit promoted posts in ML/STEM subs + YouTube pre-roll on lecture/study-method videos + X #studytwt promoted — match intent (see S9-A for exact placements)	Inf · 62%	Ritsu channels; Traffic Lasso; S9-A
13	Channel we avoid	<i>Which channels are a waste here?</i>	Intent-less mass-blasts (general feed TikTok/IG/Facebook, cold LinkedIn) waste budget — but PRECISELY-targeted placements on study/ML creators + communities DO convert (S9-A). Avoid the broad reach, not the platform	Inf · 58%	locked wedge; S9-A
14	Device for discovery	<i>Do they discover on phone, buy on laptop?</i>	Laptop/desktop — the upload-dense-PDF-then-study workflow is keyboard-and-big-screen; mobile only for video	Obs · 82%	analytics (PDF-heavy, revisit)
15	Preferred content length	<i>Short hit or deep dive?</i>	Long where it earns it: 40–90 min lectures, multi-chapter PDFs — but values the <60s magic moment as the proof hook	Obs · 80%	analytics; Ritsu spec
★16	Virality mechanism	<i>How would this spread within the segment?</i>	Share-link of a generated quiz / Knowledge Map to coursemates ("try this on the midterm material") — works mechanically; unproven on strangers	Obs · 84%	analytics (15 shares, 0 organic)

#	Attribute	Core Question	Answer	Conf	Source
17	Beta-recruit channel	<i>Where can we hand-recruit the first N testers?</i>	Hand-recruit the N=10 from r/learnmachinelearning, fast.ai forums, course Discords/Slacks, #studytwit — each tester mid-course with a live deadline (Collison-install, SOP-PRODUCT-002)	Inf · 62%	SOP-PRODUCT-002 (VI-added)
18	Creator-partnership fit	<i>Which creators could we co-build with, at what scale?</i>	High-fit, mid-scale: learning-science creators (Justin Sung ~195K, Ali Abdaal slice) + ML educators (StatQuest, Krish Naik) — authority > raw reach for this skeptical audience; sponsor/co-build, not mass-blast	Inf · 58%	S9-A (VI-added)

S9-A · Ad-Targeting Channel & Location Map (deep research)

Where the Deadline-Bearing STEM/ML Masterer actually spends attention — a workflow of 11 parallel research agents (one per platform) returned the top specific, **ad-targetable** entities for each. Each row is something you can paste into an ad manager (a channel placement, a subreddit, a handle, a group, a hashtag) or hand-seed organically. `Cat` = study-method / ml-ai / stem-course / exam-prep / general-learning. Use this directly to configure paid campaigns; for Discord/forums, treat it as the organic Collison-install recruiting map (SOP-PRODUCT-002 / SOP-CUSTOMER-006).

Quick reference — platform → ad mechanism

Platform	Top entities	Primary ad / seeding mechanism
YouTube (channels)	30	YouTube Ads — managed-placement (channel) + custom-intent + topics
Reddit (subreddits)	30	Reddit Ads — community (subreddit) + keyword targeting
X (Twitter) (accounts and hashtags)	30	X Ads — follower look-alike + keyword + handle targeting
Facebook (groups and pages)	30	Meta Ads — detailed-interest + group-adjacent + lookalike
Instagram (accounts and hashtags)	30	Meta/IG Ads — interest + creator-audience + hashtag-adjacent
TikTok (creators and hashtags)	20	TikTok Ads — interest + hashtag + Spark (creator) Ads
Discord (servers)	20	Organic seeding (Collison-install) — not paid placements
LinkedIn (groups and influencers)	18	LinkedIn Ads — group + skill + job-title(Student) + field-of-study
Podcasts (shows)	18	Host-read sponsorship slots
Newsletters / Substack (newsletters)	18	Newsletter sponsorship / classified slots
Forums & communities (forums and communities)	18	Organic seeding + community participation

Total: 262 ad-targetable entities across 11 platforms.

YouTube — channels (30)

Ad mechanism: YouTube Ads — managed-placement (channel) + custom-intent + topics.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	3Blue1Brown (Grant Sanderson) youtube.com/@3blue1brown	~8.3M subscribers	The canonical visual-intuition channel for linear algebra, calculus and neural-network math that cs231n/CS229/DeepLearning.AI students watch to actually understand what their course is teaching.	Use YouTube placement targeting (managed placements) on this exact channel for in-stream ads; also feed its video URLs into a custom-intent audience and mirror its math/neural-net keywords as topic targets.	ml-ai
2	StatQuest with Josh Starmer youtube.com/@statquest	~1.6M subscribers	The go-to channel for ML/stats students cramming for graded data-science and ML courses — exactly the 'I need to truly get this concept before the exam' mindset Ritsu serves.	Managed-placement in-stream ads on this channel; clone its 'StatQuest', 'machine learning clearly explained', 'logistic regression' video URLs into a custom-intent campaign targeting active ML learners.	ml-ai
3	Khan Academy youtube.com/@khanacademy	~9.4M subscribers	The largest STEM-course audience on YouTube — calculus, physics, chem, bio learners following along with a graded school course, the textbook-chapter-to-mastery job Ritsu automates.	Placement targeting on the channel plus its calculus/orgo/physics playlists; pair with custom-intent keywords like 'ace my calculus exam' and 'study for chem test'.	stem-course
4	The Organic Chemistry Tutor youtube.com/@TheOrganicChemistryTutor	~5M+ subscribers	Pre-med, chem and physics students binge this the week before a graded exam — peak 'dated deadline + dense material' moment where Ritsu turns the lecture PDF into active quizzing.	Managed placement on this channel; build a custom-intent audience from its exam-review video URLs ('final exam review', 'orgo reactions') to catch viewers in active cram mode.	stem-course
5	freeCodeCamp.org youtube.com/@freecodecamp	~10M+ subscribers	Engineers and bootcamp learners working through multi-hour course-length ML/CS lectures — the exact 'learn a new framework / pass the cohort' job, and they already use long-form video as study material.	Placement targeting on the channel and its full-course ML/Python/data-science uploads; complement with custom-intent keywords ('data science bootcamp', 'machine learning full course').	general-learning
6	Ali Abdaal youtube.com/@aliabdaal	~6.6M subscribers	The largest study-productivity audience — ex-medic whose viewers are exam-driven students seeking evidence-based study methods, the self-identified 'serious learner' Ritsu targets.	Placement targeting on the channel; layer custom-intent keywords ('how to study for exams', 'active recall', 'spaced repetition') to isolate the exam-prep slice of his broad audience.	study-method
7	Justin Sung youtube.com/@JustinSung	~195K subscribers	Highest-fit channel: a learning-science coach whose entire audience is committed learners trying to replace passive re-reading with active, durable methods — Ritsu's exact thesis.	Managed-placement in-stream on this channel (small but laser-targeted); his 'learn faster', 'active learning', 'study less study smart' video URLs make an ideal custom-intent seed list.	study-method
8	DeepLearning.AI (Andrew Ng) youtube.com/@DeepLearningAI	~600K subscribers	Official channel for the most popular paid ML curriculum (DeepLearning.AI / Coursera specializations) — viewers are enrolled, deadline-bearing ML course-takers by definition.	Placement targeting on the channel; build custom-intent from 'Andrew Ng', 'deep learning specialization', 'machine learning course' searches to reach learners mid-course.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
9	Andrej Karpathy youtube.com/@AndrejKarpathy	~806K subscribers	His 'Neural Networks: Zero to Hero' / 'building makemore & GPT' lectures are de-facto coursework for serious ML self-learners — long, dense video they study from, not entertainment.	Managed placement on this channel and its lecture-series videos; custom-intent seeds 'Karpathy nanoGPT', 'build GPT from scratch', 'neural networks zero to hero'.	ml-ai
10	MIT OpenCourseWare youtube.com/@mitocw	~6M subscribers	Full university lecture series (6.S191, 18.06 linear algebra, 6.034 AI) — the most-subscribed .edu channel; viewers are following rigorous, graded-equivalent STEM courses.	Placement targeting on the channel and specific course playlists (linear algebra, intro to deep learning); pair with custom-intent keywords for those exact course codes.	stem-course
11	Krish Naik youtube.com/@krishnaik06	~1.4M subscribers	End-to-end data-science/ML tutorials and roadmaps for learners working toward a job or completing a cohort — practical, course-style content for committed ML students.	Placement targeting on the channel; custom-intent from his 'machine learning roadmap', 'data science interview', 'end to end ML project' videos to catch goal-driven learners.	ml-ai
12	Two Minute Papers (Károly Zsolnai-Fehér) youtube.com/@TwoMinutePapers	~1.6M subscribers	Research-paper-breakdown channel whose audience is ML students and researchers who routinely 'break down a dense research paper' — a named Ritsu situational workflow.	Placement targeting on the channel; mirror its audience with custom-intent keywords like 'research paper explained', 'AI paper summary', 'read ML papers'.	ml-ai
13	Computerphile youtube.com/@Computerphile	~2.6M subscribers	University-faculty-led CS explainers (algorithms, ML, theory) watched by CS undergrads supplementing graded coursework — the engineer/student personas.	Placement targeting on the channel; custom-intent seeds around 'computer science explained', 'how algorithms work', 'neural network explained' for CS course-takers.	stem-course
14	sentdex (Harrison Kinsley) youtube.com/@sentdex	~1.3M subscribers	Hands-on Python/ML/neural-net tutorial series that learners follow as coursework to master deep learning — code-exercise-heavy, exactly Ritsu's engineer wedge.	Placement targeting on the channel and its 'Neural Networks from Scratch' / 'Python ML' series; custom-intent keywords 'learn machine learning python', 'neural network tutorial'.	ml-ai
15	Thomas Frank youtube.com/@Thomasfrank	~2.9M subscribers	Study-and-productivity channel built on exam-prep and note-taking systems; his viewers are students explicitly trying to study more effectively for graded courses.	Placement targeting on the channel; isolate study intent with custom-intent keywords ('study tips for finals', 'note taking method', 'how to remember what you read').	study-method
16	Professor Leonard youtube.com/@ProfessorLeonard	~730K-800K subscribers	Full-length college calculus and statistics lectures that students watch alongside a graded math course — high-intent, deadline-bearing STEM learners.	Managed placement on the channel and its Calculus 1/2/3 + statistics playlists; custom-intent keywords 'calculus exam review', 'pass calculus', 'statistics course'.	stem-course
17	Yannic Kilcher youtube.com/@YannicKilcher	~308K subscribers	Deep ML-paper walkthroughs for advanced students and grad-level learners decomposing dense research — directly the researcher persona's 'break down a paper' job.	Managed-placement in-stream on this channel; custom-intent seeds from his paper-review video URLs ('transformer paper explained', 'ML paper review') for advanced learners.	ml-ai
18	IBM Technology youtube.com/@IBMTechology	~1M+ subscribers	Concise AI/ML/LLM concept explainers used by professionals and students upskilling on machine-learning fundamentals — the professional career-changer persona.	Placement targeting on the channel; custom-intent keywords 'what is machine learning', 'LLM explained', 'AI fundamentals' to reach upskilling professionals.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
19	Gohar Khan youtube.com/@goharsguide	~600K subscribers	MIT-grad study/exam-prep coach whose audience is ambitious students optimizing for grades and admissions — high overlap with the committed-masterer ICP.	Placement targeting on the channel; custom-intent keywords 'how to get straight A's', 'study for exams', 'memorize faster' to capture exam-driven students.	exam-prep
20	Serrano.Academy (Luis Serrano) youtube.com/@SerranoAcademy	~150K subscribers	Friendly-but-rigorous ML & math explainer (neural nets, transformers, linear algebra) for learners who want to genuinely understand the math behind an ML course.	Managed-placement in-stream on this channel; seed custom-intent with 'machine learning math', 'transformers explained', 'linear algebra for ML'.	ml-ai
21	CodeEmporium (Ajay Halthor) youtube.com/@CodeEmporium	~130K subscribers	Implements transformers, GANs and LLMs in plain Python with explanation — niche but exactly the deep-learning student who needs active practice on dense material.	Managed-placement in-stream on this channel; custom-intent keywords 'transformer from scratch', 'attention is all you need explained', 'deep learning in python'.	ml-ai
22	Andrew Ng (Stanford Online / personal lectures) youtube.com/@stanfordonline	~700K subscribers (Stanford Online)	Hosts full CS229/CS230 machine-learning and deep-learning lecture courses — the literal Stanford ML coursework the persona is enrolled in or shadowing.	Placement targeting on the channel and the CS229/CS230 lecture playlists; custom-intent keywords 'CS229', 'Stanford machine learning', 'CS231n lecture'.	stem-course
23	Michel van Biezen (ilectureonline) youtube.com/@MichelmanBiezen	~1.4M subscribers	Massive library of bite-size physics, math and engineering lectures students use to master specific graded-course topics before exams.	Placement targeting on the channel and its physics/calculus playlists; custom-intent keywords 'physics exam help', 'engineering mechanics', 'circuit analysis problems'.	stem-course
24	PatrickJMT youtube.com/@patrickjmt	~1.4M subscribers	Just-Math-Tutorials channel that calculus and college-math students hit when stuck on graded homework or cramming for a test — pure deadline-driven STEM study.	Placement targeting on the channel; custom-intent keywords 'calculus help', 'derivative examples', 'math homework' to catch students mid-assignment.	stem-course
25	Veritasium (Derek Muller) youtube.com/@veritasium	~18M subscribers	Science-explainer giant whose founder's PhD is literally in physics-education research on why passive video alone doesn't teach — a values-aligned, science-curious learner audience.	Placement targeting on the channel (huge reach, broader fit); tighten with custom-intent keywords 'learn physics', 'understand science', 'why we forget what we learn' to skew to serious learners.	general-learning
26	CrashCourse youtube.com/@crashcourse	~16M subscribers	Course-aligned crash content (biology, chemistry, statistics, organic chem) that high-school and college students watch to review a graded subject under time pressure.	Placement targeting on the channel and its STEM playlists (Chemistry, Statistics, Organic Chemistry); custom-intent keywords 'crash course chemistry', 'study for biology exam'.	general-learning
27	Stanford CS231n / Stanford NLP (course channels) youtube.com/@stanfordonline (CS231n & CS224n playlists)	lecture playlists with 100K-1M+ views each	The exact named-in-brief courses (CS231n vision, CS224n NLP) the ML persona is enrolled in — highest possible intent match for deadline-bearing ML coursework.	Managed placement on the specific CS231n/CS224n lecture videos; custom-intent audience built from 'CS231n', 'CS224n', 'convolutional neural networks course' searches.	stem-course
28	Luke's Lab / Zach Star (engineering & math) youtube.com/@zachstar	~1.2M subscribers	Engineering-and-applied-math channel followed by STEM majors choosing and surviving rigorous courses — committed degree-seeking learners with graded deadlines.	Placement targeting on the channel; custom-intent keywords 'engineering math', 'how to study STEM', 'differential equations' to reach degree-seeking STEM students.	stem-course

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
29	NeetCode youtube.com/@NeetCode	~900K subscribers	CS students and bootcampers grinding algorithms for graded data-structures courses and coding-interview deadlines — high-intent, deadline-bearing, accuracy-obsessed learners.	Placement targeting on the channel; custom-intent keywords 'leetcode prep', 'data structures and algorithms', 'coding interview' to catch deadline-driven CS learners.	exam-prep
30	Gohar Khan alt / Mike Dane-style + Math Sorcerer youtube.com/@TheMathSorcerer	~700K subscribers	Channel about how to self-study rigorous math (real analysis, linear algebra) and survive proof-based courses — the committed STEM masterer who wants depth, not shortcuts.	Placement targeting on the channel; custom-intent keywords 'how to study math', 'self study linear algebra', 'real analysis' to reach proof-course STEM learners.	stem-course

Reddit — subreddits (30)

Ad mechanism: Reddit Ads — community (subreddit) + keyword targeting.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	r/learnmachinelearning reddit.com/r/learnmachinelearning/	~500K members	The single most on-persona community: people actively learning ML from courses (cs231n, Andrew Ng, fast.ai) who need to convert dense lecture material into practice.	Primary community-target on Reddit Ads. Pair with keyword targeting on 'cs231n', 'CS229', 'study plan', 'how to learn'. Lead creative: 'Turn your ML lecture PDFs into quizzes in 30 seconds.'	ml-ai
2	r/MachineLearning reddit.com/r/MachineLearning/	~3.0-3.1M members	The largest ML hub; contains many grad students and self-learners breaking down research papers into testable knowledge.	High-reach community target but lower purity (research-heavy). Constrain with keyword targeting on 'paper', 'reproduce', 'study', 'exam' and lean creative toward the 'break down a research paper' workflow rather than beginner messaging.	ml-ai
3	r/learnprogramming reddit.com/r/learnprogramming/	~4.3M members	Huge population of bootcamp/CS-course learners with graded deadlines who are the engineer/self-learner personas; learning a new framework from docs is a named Ritsu workflow.	Largest-reach community in this set. Pair with keyword targeting 'bootcamp', 'course', 'study', 'docs'. Creative angle: 'Learn a new framework fast — drop the docs in, get quizzed.' Expect broad audience, so narrow with keywords.	general-learning
4	r/GradSchool reddit.com/r/GradSchool/	~924K members	Grad students facing qualifying exams and dense reading lists — exactly the deadline-bearing, mastery-motivated learner who must internalize papers and textbooks.	Strong community target for the 'qualifying exam' and 'research paper' angles. Keyword-pair with 'quals', 'comprehensive exam', 'paper'. Creative: 'Prepping for quals? Turn your reading list into active recall.'	general-learning
5	r/premed reddit.com/r/premed/	~902K members	Pre-meds are archetypal deadline-driven masterers (MCAT, hard science courses) who already pay for Anki and study tools Ritsu replaces.	High-intent community target. Pair with keyword targeting 'Anki', 'study schedule', 'orgo', 'biochem'. Creative: 'Stop hand-making Anki decks — auto-generate from your own notes.' Note Reddit ad-policy sensitivity around exam/medical content; keep claims accuracy-led.	exam-prep

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
6	r/EngineeringStudents reddit.com/r/EngineeringStudents/	~450-800K members (sources vary; ~800K most recent)	'For those stuck doing math at 3am' — committed STEM students with graded multi-week courses (calc, physics, thermo) and looming exams.	Core STEM-course community target. Keyword-pair 'exam', 'final', 'study group', 'professor notes'. Creative: 'Ace your exam in 3 days — turn lecture slides into a study plan.' Strong fit for the 'exam in 3 days' wedge.	stem-course
7	r/GetStudying reddit.com/r/GetStudying/	~1.5M members	The largest study-method community; members are explicitly seeking better active-study techniques after being burned by passive re-reading.	Top-of-list for study-method targeting + high reach. Pair with keyword targeting 'active recall', 'spaced repetition', 'how to study'. Creative leads with the learning-science angle: 'Active recall beats re-reading — automated from any document.'	study-method
8	r/Anki reddit.com/r/Anki/	~250K members	These users have already adopted spaced repetition and feel the pain of manual card creation — the exact substitute Ritsu auto-generates from a source.	Highest-purity substitution target. Community-target + keyword 'card creation', 'deck', 'add-on'. Creative must be respectful (Anki-loyal audience): 'Love Anki, hate making cards? Auto-generate flashcards from your PDFs.' Seed organically too.	study-method
9	r/MCAT reddit.com/r/Mcat/	~170K+ members	A dated, high-stakes exam community where members live in Anki/UWorld and want genuine mastery of dense content under deadline.	Very high-intent exam-prep community target. Keyword-pair 'Anki', 'content review', 'practice questions'. Creative: 'Turn your content review into quizzes and flashcards instantly.' Keep accuracy/no-hallucination front and center for this skeptical, rigor-valuing crowd.	exam-prep
10	r/datascience reddit.com/r/datascience/	~800K-1M members	Practitioners and bootcamp learners upskilling in ML/stats from courses and papers — overlaps heavily with the ML-mastery persona.	Community target for the self-learner/upskiller angle. Pair with keyword 'course', 'certification', 'learn', 'study'. Creative: 'Master a new DS course faster — quizzes auto-built from your material.'	ml-ai
11	r/learnpython reddit.com/r/learnpython/	~1.3M members	Massive cohort of course-driven Python learners (many doing ML/data-science tracks) who need to retain concepts under deadline.	High-reach community target with a learning intent baked into the name. Keyword-pair 'course', 'tutorial', 'exercises'. Creative emphasizes code-exercise generation: 'Drop in the docs, get code exercises to practice.'	general-learning
12	r/MedicalSchool reddit.com/r/medschool/	~300K+ members	Med students face relentless dated exams (Step 1/2, shelf) with enormous reading loads — the deadline-bearing masterer at maximum intensity.	Strong exam-prep community target. Keyword-pair 'Step 1', 'shelf', 'Anki', 'First Aid'. Creative: 'Turn lecture slides into active recall before your next shelf.' Maintain accuracy-led messaging for this rigor-obsessed audience.	exam-prep
13	r/deeplearning reddit.com/r/deeplearning/	~180-200K members	Focused DL learners working through courses (cs231n, fast.ai) and papers who want to truly understand, not just skim.	Precise ml-ai community target. Keyword-pair 'cs231n', 'course', 'paper', 'study'. Creative leans into the Knowledge Map / concept-connection feature: 'See how DL concepts connect — from your own lecture notes.'	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
14	r/college reddit.com/r/college/	~221K members	General undergrad community full of students with graded multi-week courses and finals — broad reach into the student persona.	Mid-reach community target for the general student wedge. Keyword-pair 'finals', 'midterm', 'study', 'GPA'. Creative: 'Finals in 3 days? Turn your slides into a study plan.' Use as a volume complement to the precise STEM subs.	general-learning
15	r/learnmath reddit.com/r/learnmath/	~203K members	Students grinding through calculus/linear algebra/stats coursework who need to convert dense material into practice problems.	STEM-course community target. Keyword-pair 'calculus', 'linear algebra', 'exam', 'practice problems'. Creative: 'Turn your math textbook chapter into practice problems and a concept map.'	stem-course
16	r/MLQuestions reddit.com/r/MLQuestions/	~70-90K members	A help-seeking ML-learner community — people stuck mid-course who want to genuinely understand, the exact mindset Ritsu serves.	Smaller but high-purity ml-ai community target. Keyword-pair 'understand', 'course', 'confused', 'explain'. Creative leans on tutoring modes: 'A tutor that quizzes you and explains concepts back — from your own material.'	ml-ai
17	r/statistics reddit.com/r/statistics/	~600K members	Stats coursework underpins ML and many STEM degrees; members face graded problem sets and exams in a notoriously dense subject.	Community target for the STEM-course angle. Keyword-pair 'exam', 'homework', 'course', 'understand'. Creative: 'Finally understand stats — quizzes and explanations from your own lecture notes.'	stem-course
18	r/AskStatistics reddit.com/r/AskStatistics/	~120K members	Help-seeking stats students mid-course who want conceptual mastery — high willingness to try a tool that explains and tests them.	Niche, high-intent community target. Keyword-pair 'concept', 'explain', 'course', 'exam'. Creative: 'Stuck on a stats concept? Get it explained, then quiz yourself on it.'	stem-course
19	r/calculus reddit.com/r/calculus/	~140K members	Single-subject community of students in dated calculus courses with frequent exams — precisely on the STEM-course wedge.	Highly precise STEM-course community target. Keyword-pair 'exam', 'final', 'practice', 'help'. Creative: 'Calc final this week? Turn your textbook into practice problems.' Great low-CPM, high-purity test buy.	stem-course
20	r/chemhelp reddit.com/r/chemhelp/	~23K members	Organic/general chemistry students (incl. pre-meds) in dated courses needing active practice on a famously hard subject.	Small but pure STEM-course (orgo) target. Keyword-pair 'orgo', 'mechanism', 'exam', 'practice'. Creative: 'Master orgo mechanisms — quizzes built from your own slides.' Best paired with r/premed for the chem-heavy audience.	stem-course
21	r/artificial reddit.com/r/artificial/	~1M+ members	Broad AI-interested audience that includes many self-learners actively studying ML; useful for top-of-funnel reach with keyword narrowing.	High-reach but low-purity community target — MUST narrow with keyword targeting 'learn', 'course', 'study', 'tutorial'. Creative: 'Actually learn AI (not just read about it) — quizzes from any source.'	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
22	r/studytips reddit.com/r/studytips/	~11K members	Small but exact study-method community where members are explicitly hunting better, more active ways to learn.	Low-volume, high-purity study-method target. Keyword-pair 'active recall', 'method', 'spaced repetition'. Creative is the learning-science manifesto: 'The active-study method that sticks — automated.' Good for organic seeding too.	study-method
23	r/Professors reddit.com/r/Professors/	~340K members	Educators are persona #5 (creating courses/materials for students); this is where instructors discuss teaching tools and content creation.	Community target for the educator angle. Keyword-pair 'course materials', 'quiz', 'assignment'. Creative: 'Build a course for your students from existing material.' Distinct messaging from the student-facing buys.	general-learning
24	r/gradadmissions reddit.com/r/gradadmissions/	~250K members	Prospective grad students prepping for GRE/subject tests and brushing up on coursework — deadline-bearing exam-prep mindset.	Community target for the exam-prep/transition angle. Keyword-pair 'GRE', 'subject test', 'prep', 'study'. Creative: 'Prepping for the GRE subject test? Turn prep books into active recall.'	exam-prep
25	r/computerscience reddit.com/r/computerscience/	~2.4M members	CS students and learners working through theory-heavy courses (algorithms, ML) with exams — broad reach into the STEM-CS persona.	Large-reach STEM/CS community target; narrow with keyword 'exam', 'course', 'algorithms', 'study'. Creative: 'Turn your algorithms lecture notes into a quiz before the exam.'	stem-course
26	r/PhysicsStudents reddit.com/r/PhysicsStudents/	~180K members	Physics students in dated, problem-set-heavy courses who need active practice and concept mastery — squarely on the STEM-course wedge.	Precise STEM-course community target. Keyword-pair 'exam', 'problem set', 'midterm', 'help'. Creative: 'Physics midterm coming? Turn your notes into practice problems and a concept map.'	stem-course
27	r/Mcat (study-tools crossover) / r/step1 reddit.com/r/step1/	~90K members	USMLE Step 1 examinees are peak deadline-driven masterers drowning in Anki and question banks Ritsu can auto-generate.	High-intent exam-prep community target. Keyword-pair 'Anki', 'question bank', 'First Aid', 'dedicated'. Creative: 'Auto-generate practice questions from your First Aid notes.' Accuracy-led claims essential.	exam-prep
28	r/Mathematics reddit.com/r/mathematics/	~1M members	Math majors and serious learners in proof/analysis courses who must internalize dense material — broad reach into the STEM-math persona.	Large-reach community target; narrow with keyword 'course', 'exam', 'proof', 'study'. Creative: 'Master your analysis course — quizzes and concept maps from your textbook.'	stem-course
29	r/MLQuestions (peer) / r/AskAcademia reddit.com/r/AskAcademia/	~250K members	Grad students and researchers discussing how to digest papers and prep for exams/teaching — overlaps the researcher and grad-learner personas.	Community target for the researcher/grad angle. Keyword-pair 'paper', 'reading', 'qualifying exam', 'teaching'. Creative: 'Break down a dense research paper into testable knowledge.'	general-learning
30	r/Notion reddit.com/r/Notion/	~400K members	Students using Notion for note storage are the passive-study substitute Ritsu replaces; high overlap with method-seeking learners.	Substitution-angle community target. Keyword-pair 'student', 'study', 'notes', 'flashcards'. Creative: 'Notes aren't studying — turn your Notion notes into active quizzes.' Position as a complement-then-replacement, not an attack.	study-method

X (Twitter) — accounts and hashtags (30)

Ad mechanism: X Ads — follower look-alike + keyword + handle targeting.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	Andrew Ng (account) @AndrewYNg — https://x.com/AndrewYNg	~1.1M followers	The single most-followed figure for exactly this persona — DeepLearning.AI / Coursera / Stanford CS229 learners follow him while taking his courses with graded deadlines.	Use as a follower look-alike seed (his audience IS enrolled ML learners); also handle-target @AndrewYNg directly so ads serve to his engagers and replies.	ml-ai
2	Andrej Karpathy (account) @karpathy — https://x.com/karpathy	~1.4M followers	His 'Zero to Hero' / nn lectures are the de-facto self-study ML curriculum; his followers are the deadline-driven masterers turning his videos into practice.	Follower look-alike seed (highest-intent ML-learner audience on X) plus handle-targeting; pair with keyword targeting on 'makemore', 'micrograd', 'nanoGPT'.	ml-ai
3	Lex Fridman (account) @lexfridman — https://x.com/lexfridman	~4.2M followers	Largest AI/ML-adjacent audience on X; heavily skews toward students and self-learners aspiring to master AI, a broad top-of-funnel pool for the wedge.	Use as a broad follower look-alike seed for reach; layer with ML-keyword targeting to filter his general audience down to active learners.	ml-ai
4	Santiago (svpino) (account) @svpino — https://x.com/svpino	~383K followers	Self-describes as teaching 'hard-core Machine Learning'; his audience is practitioners and students actively building ML skill against real goals.	Strong follower look-alike seed — tightly on-persona (ML learners, not just AI news). Handle-target his engagers for high relevance.	ml-ai
5	3Blue1Brown / Grant Sanderson (account) @3blue1brown — https://x.com/3blue1brown	~418K followers	The canonical visual-math channel for linear algebra/calculus/neural-nets — exactly the STEM foundations this persona studies for ML and engineering courses.	Follower look-alike seed to reach math/STEM masterers; ideal for the 'turn a dense math lecture into active practice' ad angle.	stem-course
6	Yann LeCun (account) @ylecun — https://x.com/ylecun	~900K+ followers	Foundational deep-learning figure followed by serious ML students and grad researchers who self-identify as wanting to master the field rigorously.	Follower look-alike seed for the rigor-minded ML grad/research audience; combine with @GoogleDeepMind for an ML-research interest cluster.	ml-ai
7	François Chollet (account) @fchollet — https://x.com/fchollet	~450K+ followers	Keras creator and ARC author; his audience is hands-on ML learners and engineers learning deep learning from his book/framework.	Follower look-alike seed; keyword-target 'Keras', 'deep learning with python' to catch learners working through his materials.	ml-ai
8	Jeremy Howard (account) @jeremyphoward — https://x.com/jeremyphoward	~250K+ followers	Co-founder of fast.ai — a named archetype course in the persona definition; his followers are mid-course practical-DL learners with concrete goals.	High-precision follower look-alike seed (fast.ai cohort); handle-target plus keyword 'fastai' to reach learners actively in the course.	ml-ai
9	Sebastian Raschka (account) @rasbt — https://x.com/rasbt	~290K+ followers	Author of 'Build a Large Language Model (from Scratch)' and ML textbooks; his audience learns by implementing — the active-mastery mindset Ritsu sells.	Follower look-alike seed tightly aligned to learn-by-doing ML students; keyword-target 'from scratch', 'LLM' to reach implementers.	ml-ai
10	Tivadar Danka (account) @TivadarDanka — https://x.com/TivadarDanka	~150K followers	Author of 'Mathematics of Machine Learning'; his entire audience is people grinding linear algebra/calculus/probability to master ML — bullseye persona.	Very high-precision follower look-alike seed (math-for-ML learners); perfect for the 'master the math behind ML' creative.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
11	Mark Tenenholtz (account) @marktenenholtz — https://x.com/marktenenholtz	~250K+ followers	Kaggle Master and data-science educator whose audience is upskilling data-science/ML learners and bootcamp-style practitioners.	Follower look-alike seed for the data-science learner segment; pair with @svpino and @rasbt to build an ML-educator audience cluster.	ml-ai
12	Anton (abacaj) (account) @abacaj — https://x.com/abacaj	~110K+ followers	Hands-on LLM/ML engineer whose followers are practitioners learning to build and fine-tune models — active, deadline-agnostic but high-intent learners.	Follower look-alike seed for applied-LLM learners; keyword-target 'fine-tune', 'LLM' to catch the build-to-learn cohort.	ml-ai
13	Rachel Thomas (account) @math_rachel — https://x.com/math_rachel	~80K+ followers	fast.ai co-founder focused on practical, accessible ML/math education; her audience is learners (esp. via courses) tackling deep learning rigorously.	Follower look-alike seed overlapping the fast.ai learner base; combine with @jeremyphoward for a course-cohort audience.	ml-ai
14	Google DeepMind (account) @GoogleDeepMind — https://x.com/GoogleDeepMind	~1M+ followers	Followed by aspiring and current ML grad students/researchers who want to master the field at a research level — the rigor end of the persona.	Broad follower look-alike seed for ML-research aspirants; filter with course/exam keywords to reach those actively studying vs. passive fans.	ml-ai
15	Sebastian Ruder / NLP-research-adjacent learners (account) @seb_ruder — https://x.com/seb_ruder	~120K+ followers	NLP researcher whose followers are grad students and self-learners mastering ML/NLP from papers — directly matches the 'break down a research paper' job.	Follower look-alike seed for the paper-reading ML learner; ideal for the 'turn a research paper into a quiz' ad angle.	ml-ai
16	Patrick Loeber (account) @patloeber — https://x.com/patloeber	~71K followers	Python/ML tutorial creator (DeepMind DevRel) whose audience is beginner-to-intermediate ML learners following structured lessons — high active-learning intent.	Follower look-alike seed for tutorial-following ML learners; pair with keyword 'pytorch', 'python machine learning'.	ml-ai
17	Dan Kornas (account) @dankornas — https://x.com/dankornas	~87K followers	Bio 'Building the best AI learning resource' — his entire audience is people explicitly trying to learn/master AI, a clean persona match.	Follower look-alike seed of self-identified AI learners; strong fit for a head-to-head 'replace your study stack' creative.	ml-ai
18	Cal Newport (account) @calnewport — https://x.com/calnewport	~180K+ followers	CS professor and 'Deep Work' / 'How to Become a Straight-A Student' author; his audience is serious students optimizing study and focus for hard courses.	Follower look-alike seed for the study-method/learning-science segment; ideal for the 'active learning beats passive re-reading' angle.	study-method
19	Ali Abdaal (account) @AliAbdaal — https://x.com/AliAbdaal	~250K+ followers	Ex-doctor, #1 productivity/study creator whose audience is students seeking evidence-based study techniques (active recall, spaced repetition) for exams.	Broad follower look-alike seed for the study-method audience; layer student/exam keywords to focus on the in-course subset.	study-method
20	Justin Sung (account) @justinsungofficial — https://x.com/justinsungofficial	~30K+ followers (X); 1M+ across platforms	The '#1 studying coach' focused on active learning and effective encoding — his audience is learners deliberately upgrading how they study for hard material.	Follower look-alike seed precisely on the learning-science angle; perfect for creative tying Ritsu to active-recall/effective-learning science.	study-method

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
21	#100DaysOfMLCode (hashtag + hub account) #100DaysOfMLCode — hub: @100DaysOfMLCode https://x.com/100daysofmlcode	Tens of thousands of participants; active retweet hub account	A public daily-commitment ML-learning challenge — literally deadline-driven self-learners logging ML study, the persona in motion.	Keyword/hashtag-target #100DaysOfMLCode to serve ads to people actively tweeting their ML study logs; handle-target the @100DaysOfMLCode hub's engagers.	ml-ai
22	#studytwt (hashtag) #studytwt	Large active student community (hundreds of thousands of tweets/users)	The core study-Twitter community where students post exam-prep, notes and motivation — the deadline-bearing learner's home base.	Keyword-target 'studytwt' plus 'exam', 'finals', 'study with me' to reach students mid-grind; best for general-mastery and exam-prep creatives.	general-learning
23	#MachineLearning (hashtag) #MachineLearning	Very high volume — millions of tweets; broad ML audience	Broadest ML topic tag capturing learners, students and practitioners discussing and studying ML across all levels.	Keyword-target #MachineLearning for reach, then narrow with 'learn', 'course', 'study' modifiers or layer an educator follower look-alike to filter to learners.	ml-ai
24	#AcademicTwitter (hashtag) #AcademicTwitter	Large global community of grad students, researchers and academics	Home of grad students and researchers facing quals and dense papers — directly matches the 'break down a research paper / pass quals' persona slice.	Keyword-target #AcademicTwitter and #gradschool for the research-paper-to-quiz angle; pair with STEM-field keywords to focus on technical fields.	general-learning
25	#datascience (hashtag) #datascience	Very high volume — millions of tweets; large practitioner+learner base	Captures bootcamp students and self-learners upskilling in data science/ML who turn courses and notebooks into practice.	Keyword-target #datascience with learner modifiers ('learning', 'bootcamp', 'studying'); combine with @marktenenholtz/@svpino look-alikes to focus on learners.	ml-ai
26	#gradschool (hashtag) #gradschool	Large active community of graduate students	Grad students juggling qualifying exams, hard coursework and dense reading — high willingness to pay to master material before a graded deadline.	Keyword-target #gradschool + #PhD + 'quals' for the exam-prep angle; serve the 'ace your exam / break down a paper' creative.	exam-prep
27	fast.ai (account) @fastdotai — https://x.com/fastdotai	~80K+ followers	Official account of a named archetype course; followers are enrolled/active practical-deep-learning students working through a structured curriculum.	High-precision follower look-alike seed (a literal course cohort); handle-target @fastdotai engagers for in-course learners.	stem-course
28	DeepLearning.AI (account) @DeepLearningAI — https://x.com/DeepLearningAI	~400K+ followers	Andrew Ng's education org; its followers are people enrolled in or shopping its ML/AI courses with assignments — the core paid-course wedge.	Strong follower look-alike seed of active ML-course students; handle-target for engagers; pair with @AndrewYNg to build a course-learner audience.	stem-course
29	Khan Academy (account) @khanacademy — https://x.com/khanacademy	~1M+ followers	Massive audience of students studying calculus, physics, chemistry and CS — the STEM-course foundations layer of the persona, broad top-of-funnel.	Broad follower look-alike seed for STEM students; narrow with course/exam keywords to reach those facing graded deadlines, not casual browsers.	stem-course

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
30	#100DaysOfCode (hashtag) #100DaysOfCode	Very large — one of X's most active learning hashtags (millions of tweets, dedicated retweet bot)	The broader public coding-learning commitment community; many participants are CS students and ML learners logging daily study against a 100-day deadline.	Keyword/hashtag-target #100DaysOfCode for high-volume reach of committed self-learners; layer ML keywords to filter toward the CS/ML subset.	general-learning

Facebook — groups and pages (30)

Ad mechanism: Meta Ads — detailed-interest + group-adjacent + lookalike.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	Artificial Intelligence & Deep Learning (DeepNetGroup) facebook.com/groups/DeepNetGroup/	~500K members	The single largest AI/ML learning group on Facebook, packed with learners working through deep-learning courses, papers, and dense lecture material — exactly Ritsu's ML-masterer wedge.	Use as a saved-audience seed: layer detailed-interest 'Deep learning' + 'Artificial intelligence' and run group-adjacent placement; build a lookalike from anyone who clicks. Optionally request an organic seed post pinned by an admin since it's news/research heavy.	ml-ai
2	Artificial Intelligence and Machine Learning facebook.com/groups/ai.ml.bigdata/ (Artificial intelligence, Machine learning, Big data Learners)	~170K members	A beginner-friendly, resource-sharing AI/ML group whose members are actively learning and asking 'how do I master this?' — the deadline-bearing self-learner archetype.	Detailed-interest target 'Machine learning' + 'Artificial intelligence (marketing)' narrowed to behaviors 'Engaged Shoppers'/students; this group's 'beginner learners' framing maps to the 'turn lecture PDFs into practice' ad angle. Seed with a free-upload demo post.	ml-ai
3	Data Mining / Machine Learning / Artificial Intelligence facebook.com/groups/1939366696159868 (Data Science - Machine Learning, Artificial intelligence)	~125K-130K members	Research-and-learning-resource oriented group — members process papers and dense ML content, the exact source material Ritsu turns into active practice.	Pair detailed-interest 'Data science' + 'Machine learning' with the 'break down a research paper into a quiz' creative. Use group membership as an audience-insight signal to refine a lookalike seeded from paper-to-quiz landing-page visitors.	ml-ai
4	Python Machine Learning & Deep Learning facebook.com — search 'Python Machine Learning & Deep Learning' (one of the most active ML groups, ~85K+)	~85K members	Highly active group of practitioners learning ML via Python — engineers learning a new framework from docs, a named Ritsu persona archetype.	Detailed-interest 'Python (programming language)' + 'Deep learning'; lead creative with the 'learn a new framework fast / code exercises from docs' angle. Strong lookalike seed for the engineer sub-persona.	ml-ai
5	Kaggle (official Page) facebook.com/kaggle/	~74K followers	Official page of the world's largest data-science community; followers are competition-driven learners who constantly upskill on graded leaderboards (a deadline analog).	Detailed-interest target 'Kaggle' directly (it is an available interest) + 'Data science'; run a lookalike off page-engagers. Creative: 'turn the course behind your next competition into active recall.'	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
6	DeepLearning.AI (official Page) facebook.com/DeepLearningAI_official (search 'DeepLearning.AI'); 7M+ learners across its courses; 7M+ learners across its courses)	7M+ learners (course base); large FB following	Andrew Ng's brand — its followers are enrolled in dated, graded DeepLearning.AI / Coursera specializations, the precise 'committed STEM/ML masterer with a deadline' wedge.	Detailed-interest target 'DeepLearning.ai' / 'Andrew Ng' / 'Coursera' as available interests; this is the highest-intent seed for the ML-course persona. Creative: 'finish your DeepLearning.AI specialization — quiz yourself on every lecture.'	ml-ai
7	Coursera (official Page) facebook.com/Coursera/	~4M+ followers	Followers are actively enrolled in dated, graded online courses (the deadline driver) across STEM/ML/data — a broad-but-on-job audience that uses passive video and needs active practice.	Detailed-interest 'Coursera' is a large available interest — use as a top-of-funnel reach audience, narrowed with 'Studying' + 'Online learning'. Creative: 'turn your Coursera lectures into quizzes that make it stick.'	general-learning
8	fast.ai (Practical Deep Learning community) facebook.com — search 'fast.ai'; community around course.fast.ai (Jeremy Howard)	Tens of thousands (course community)	fast.ai learners work through a hard, dense, multi-lesson deep-learning course on a self-imposed schedule — high-rigor, accuracy-valuing masterers.	Detailed-interest 'Deep learning' + 'Jeremy Howard'/'fast.ai' where available; narrow to engaged-with-online-courses. Creative: 'master every fast.ai lesson — active recall on the notebooks and lectures.'	ml-ai
9	MCATBros (MCAT Preparation Study Group + Premed + Med Application) facebook.com/groups/mcatbros/	~40K+ members	Premeds grinding for a single high-stakes dated exam (the MCAT) — the purest 'looming graded deadline + burned by passive study' persona, heavy Anki/Quizlet users.	Detailed-interest 'MCAT' + 'Pre-medical' + 'Anki'; build a lookalike from MCAT-landing-page visitors. Creative: 'ace your MCAT — turn your prep PDFs and lecture videos into quizzes and flashcards in 30 seconds.'	exam-prep
10	MCAT Self Prep Study Group facebook.com/groups/318062205385017/	Thousands (growing, active)	Self-directed MCAT students explicitly building their own study systems from materials — exactly the 'I want to master this myself, not be handed a course' Ritsu user.	Target detailed-interest 'MCAT' + 'Studying'; seed organically with an admin-approved post showing the upload-to-quiz flow. Lookalike off engaged clickers from the MCAT creative.	exam-prep
11	Jack Westin (MCAT) Community Group facebook.com — search 'Jack Westin' group (the home for MCAT prep + courses)	Tens of thousands	A large MCAT-prep community tied to a course brand; members are mid-prep on a dated exam and share/seek study resources daily.	Detailed-interest 'MCAT' + 'Jack Westin' (brand interest where available); creative emphasizes accuracy/'won't hallucinate' to address the skeptical-of-AI premed. Seed via group post + retarget visitors.	exam-prep
12	GMAT Club (official Page) facebook.com/gmatclubforum/	~200K+ page likes (1M+ member community overall)	The hub for GMAT/GRE test-takers preparing for a dated, high-stakes graded exam — disciplined masterers who replace flashcard/quiz tools.	Detailed-interest target 'GMAT' + 'Graduate Management Admission Test' + 'Studying'; lookalike from GMAT-prep page engagers. Creative: 'turn your GMAT prep books into active drills.'	exam-prep
13	GMAT Preparation (Worldwide) facebook.com/groups/gmatworldwide/	Tens of thousands of members	Active global GMAT-prep group; members face a fixed test date and constantly trade study methods and materials.	Detailed-interest 'GMAT' narrowed to English-speaking countries; this group is a seeding target for an organic 'paste your GMAT notes, get a quiz' demo. Build lookalike off the resulting clicks.	exam-prep

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
14	GRE / GMAT Prep Study Group (Prepez) facebook.com/groups/PrepezStudyGroup/	Tens of thousands of members	Combined GRE+GMAT prep community — grad-school applicants on a deadline who need efficient active recall over passive review.	Detailed-interest 'GRE' + 'GMAT' + 'Graduate school'; creative leads with 'exam in 3 days? turn any prep PDF into a quiz.' Seed organically; retarget engagers into a lookalike.	exam-prep
15	Programming (C, C++, C#, Java, Python, Android, Web) facebook.com — search 'Programming' group (~407K members)	~407K members	Massive group of CS learners and students working through programming coursework and docs — Ritsu's 'engineer learning a framework / code exercises' use case.	Detailed-interest 'Computer programming' + 'Computer science' narrowed to students/'College student'; high-volume reach audience. Creative: 'learn a new framework from the docs — get code exercises auto-generated.'	stem-course
16	Python Programming Beginners facebook.com/groups/pythonprogrammingbeginners/	~246K members	Huge group of self-learners and students learning Python from courses/textbooks — feeds the engineer + self-learner personas who need active practice from dense docs.	Detailed-interest 'Python (programming language)' + 'Learning to code'; narrow to 18-34 students. Creative: 'turn the Python course you're taking into quizzes and code drills.' Strong lookalike seed.	stem-course
17	Self-Taught Programmers facebook.com/groups/selftaughtprogrammers/	~140K members	Self-directed CS learners without a degree, working through bootcamp/course material on their own timeline — the committed self-learner who values genuine mastery.	Detailed-interest 'Computer programming' + 'Online courses (education)' + behavior 'Engaged Shoppers'; creative emphasizes 'master it yourself, AI is leverage not a shortcut.' Seed post + retarget.	stem-course
18	Computer Programming (C, C++, C#, Python, Java, Android) facebook.com — search 'Computer Programming' group (~96K members)	~96K members	Active CS-coursework group; members are students and learners grinding multi-week graded programming courses, a STEM-course-with-deadline match.	Detailed-interest 'Computer science' + 'Programming'; layer 'College student' life-event/behavior. Creative: 'exam or assignment due? turn your CS lecture slides into a quiz.'	stem-course
19	The Organic Chemistry Tutor (Page) facebook.com — search 'The Organic Chemistry Tutor' (companion to the 7M+-subscriber YouTube channel)	Hundreds of thousands of followers	Followers are STEM students cramming organic chemistry, physics, and calculus — high-failure-rate weed-out courses with hard graded deadlines, a core STEM-course wedge.	Detailed-interest 'Organic chemistry' + 'Physics' + 'Calculus' + 'The Organic Chemistry Tutor' (brand interest where available); lookalike off engagers. Creative: 'ace your orgo exam — turn lecture videos and textbook chapters into practice.'	stem-course
20	Data Science (largest data-science discussion group) facebook.com — search 'Data Science' (described as the largest data-science discussion community on FB)	100K+ members	Broad data-science learning community; members continuously upskill from courses, books, and papers and rely on Quizlet/Anki/ChatGPT today.	Detailed-interest 'Data science' as a large reach audience, narrowed with 'Online learning' + 'Studying'. Creative: 'replace your Anki + ChatGPT study stack — upload, get a tutor.' Seed lookalike from converters.	ml-ai
21	Artificial intelligence, Machine learning, Big data Learners facebook.com/groups/ai.ml.bigdata/	Tens of thousands of members	Explicitly a 'Learners' group — members self-identify as people studying AI/ML, the exact serious-learner identity Ritsu targets.	Detailed-interest 'Machine learning' + 'Big data' + 'Artificial intelligence'; the 'Learners' framing makes the 'true mastery, not passive watching' creative resonate. Seed + lookalike.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
22	Deep Learning, Machine Learning & Neural Networks facebook.com/groups/210735762753395/	Tens of thousands of members	Focused neural-networks/deep-learning study group; members work through hard technical material (cs231n-style) and value rigor and accuracy.	Detailed-interest 'Neural network' + 'Deep learning'; narrow to graduate students. Creative aimed at the cs231n/CS229 archetype: 'turn your lecture notes and papers into a tutor that quizzes you.'	ml-ai
23	Premed Students / Pre-Medical Community groups facebook.com — search 'Premed Students' / 'Pre-Med' (multiple large groups, tens of thousands each)	Tens of thousands per group	Premeds juggling orgo, biochem, physics and the MCAT — the densest concentration of deadline-bearing STEM masterers who buy study tools.	Detailed-interest 'Pre-medical' + 'Medical school' + 'Studying'; layer 'College student'. Creative: 'orgo, biochem, MCAT — upload any chapter, get quizzed.' Highest-WTP exam-prep seed.	exam-prep
24	Anki / spaced-repetition study communities facebook.com — search 'Anki' (multiple groups; e.g. med-student Anki communities run 10K-13K+)	10K-13K+ per group (multiple groups)	Members already practice active recall with Anki — the exact substitute Ritsu replaces by auto-generating flashcards from any source in seconds.	Detailed-interest 'Anki' is an available interest — target it directly as a high-intent study-method audience. Creative: 'stop hand-making Anki cards — upload your PDF, get the deck instantly.' Pure substitution angle.	study-method
25	Quizlet (official Page) facebook.com/quizlet/	~1M+ followers	Quizlet's audience is students who already pay for flashcard/quiz study — Ritsu's direct substitute, so its followers are pre-qualified active-study buyers.	Detailed-interest 'Quizlet' as a large available interest, narrowed to 'College student' + STEM interests. Creative: 'Quizlet, but it builds the set from your actual lecture PDF.' Direct competitor-substitution targeting.	study-method
26	Beginners, let's start their programming journey from here facebook.com — search the group name (~91K members; Python/JS/C/ game dev)	~91K members	Entry-stage coders working through programming courses on a schedule — the self-learner/engineer who needs material turned into active drills.	Detailed-interest 'Learning to code' + 'Python'; narrow to 18-30. Creative: 'learning to code from a course? get code exercises and quizzes auto-made.' Seed + lookalike from clickers.	stem-course
27	Studying / Study Motivation interest communities facebook.com — target detailed-interest 'Studying' (broad) + study-motivation Pages/groups	Millions (interest-level)	Captures the broad 'serious learner who studies hard' identity across all courses, useful for reach + lookalike scaling beyond niche ML/exam groups.	Use Facebook's 'Studying' detailed-interest as a broad reach layer, narrowed (AND) with 'Online learning' + a specific STEM/exam interest so it stays on-persona. Primary creative: 'Turn Raw Materials Into True Mastery.'	general-learning
28	edX (official Page) facebook.com/edx/	~2M+ followers	Followers are enrolled in dated, graded university-grade STEM/CS courses (MITx, HarvardX) — the committed-masterer-with-deadline audience using passive video.	Detailed-interest 'edX' + 'MOOC' + 'Online learning'; narrow with a STEM interest. Creative: 'finish your edX course — quiz yourself on every module.' Good complement to Coursera reach.	general-learning
29	Machine Learning (general open group) facebook.com — search 'Machine Learning' (multiple open groups, tens of thousands+)	Tens of thousands of members	Generic high-traffic ML group catching learners across courses (Stanford CS229, MIT 6.S191) who need dense lectures converted to active practice.	Detailed-interest 'Machine learning' as the core ML reach audience; narrow with 'Stanford University'/'Massachusetts Institute of Technology' interests to reach the named-course archetypes. Seed + lookalike.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
30	Engineering Students / University course groups facebook.com – search '[University] Class of 20XX' + 'Engineering Students' groups (tens of thousands each)	Tens of thousands per group	University course/cohort groups concentrate undergrads in calculus, physics, linear algebra, and CS with hard graded deadlines — the STEM-course wedge at scale.	Target detailed-interest 'Engineering' + 'College student' (school/major-level where available) and 'Studying'; localize creative to exam season. Creative: 'midterms week — upload your lecture slides, get a quiz.'	stem-course

Instagram — accounts and hashtags (30)

Ad mechanism: Meta/IG Ads — interest + creator-audience + hashtag-adjacent.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	#studygram instagram.com/explore/tags/studygram/	~12M+ posts (core studygram tag)	The single largest gathering point for serious self-identified students sharing active study workflows — the exact identity our deadline-bearing masterer adopts.	Seed Reels with this + sibling tags for organic reach; for paid, target Meta interest 'Study skills'/'Test (assessment)' and place Reels ads against studygram-adjacent content. Build an engagement custom audience from Reels using this tag.	study-method
2	#studymotivation instagram.com/explore/tags/studymotivation/	~40M+ posts; 35% engagement rate	High-intent tag used by learners pushing through hard, deadline-driven study — the emotional moment Ritsu sells into (passive study burnout).	Organic seeding tag with strong engagement. For paid, pair with interest 'Motivation' + 'Student' and run a 'beat the deadline with active recall' hook video; retarget tag-engagers via custom audience.	study-method
3	Med School Bro (Dr. Jacob Portnoff) instagram.com/medschoolbro/	~2M followers	Largest premed/MCAT study-tips creator in the US — audience is exactly deadline-bearing pre-professional STEM masterers studying dense material.	Top-tier creator-partnership / branded-content target (whitelist a Reel demoing 'upload your Anki deck source → quiz'). For Ads Manager, build a lookalike from his engagers or target interest 'Medical College Admission Test (MCAT)'.	exam-prep
4	Ali Abdaal instagram.com/aliabdaal/	~1M followers (8M cross-platform)	The reference 'evidence-based study technique' creator (active recall, spaced repetition) — his audience already believes active beats passive, Ritsu's core thesis.	Flagship influencer-partnership target; his active-recall framing maps 1:1 to Ritsu. In Ads Manager use interest 'Productivity' + 'Study skills' and a lookalike off his engager custom audience.	study-method
5	Physics Wallah (PW) instagram.com/physicswallah/	~4M followers	Largest Education creator globally; STEM exam-prep audience (physics/chem/math) cramming for graded exams — on-persona by subject and urgency.	India-heavy audience (secondary market for Ritsu, not first-100 US wedge) — use as an existing Meta interest 'Physics Wallah' for cheap STEM-exam reach, but cap budget vs US-led targets.	exam-prep
6	MedStudentNotes instagram.com/med_student_notes/	~394K followers	High-yield medical study notes 'by doctors for students' — audience converts dense material into study artifacts, precisely Ritsu's job-to-be-done.	Branded-content / collab target (demo 'turn this lecture PDF into flashcards'). Ads Manager: interest 'Medical school' + 'Study skills'; lookalike from a Reels engagement audience built off medical-notes content.	general-learning

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
7	Khan Academy instagram.com/khanacademy/	~313K followers	Broad STEM-course learner base (calculus, physics, chem, CS) — the canonical 'I'm learning a hard subject' interest that Ritsu turns into active practice.	Exists as a Meta ad interest ('Khan Academy') — direct interest-targeting lever for STEM learners. Pair with 'AP exams' / 'Calculus' interests for course-specific reach.	general-learning
8	ML India instagram.com/mlindia.ai/	~250K+ followers	Largest ML practitioner/enthusiast community on IG — members are actively upskilling in ML, the AI half of Ritsu's STEM/ML wedge.	Community-account collab + Reels placement against ML content. Ads Manager: interest 'Machine learning' + 'Deep learning'; build lookalike from engagers. Geo-skew to US for first-100 wedge.	ml-ai
9	Data Science Machine Learning AI instagram.com/data_science_learn/	~244K followers	Large account dedicated to simplifying ML/data-science for learners — audience is mid-course ML masterers needing active practice on papers/notebooks.	Branded-content target for a 'paper → quiz in 30s' Reel. Ads Manager interests 'Data science' + 'Artificial intelligence'; retarget account-engagers via lookalike.	ml-ai
10	datascienceinfo instagram.com/datascienceinfo/	~210K followers	AI/ML learning-focused account whose followers are actively studying to stay current in AI — strong ML-wedge fit.	Collab/whitelist target. Ads Manager: interest 'Machine learning' + 'Python (programming language)'; lookalike off engagement audience. US geo-filter for wedge.	ml-ai
11	Codingmermaid.ai (Danica Simic) instagram.com/codingmermaid.ai/	~157K followers	Data-scientist creator sharing AI/DS study + tech tips to an aspiring-practitioner audience — credible ML-educator with a learner following.	Creator-partnership target (relatable 'how I study ML' angle). Ads Manager: lookalike from her engagers + interest 'Data science'. Good for authentic Reels seeding.	ml-ai
12	med school coach instagram.com/medschoolcoach/	~101K followers	Premed advising + MCAT study tips brand — audience is deadline-bearing pre-meds mastering dense science for a high-stakes graded exam.	Brand-collab + interest target ('MCAT', 'Pre-medical'). Whitelist a Reel showing MCAT-content → active recall. Lookalike from engagers for US premed reach.	exam-prep
13	White Coats and Corgis instagram.com/whitecoatsandcorgis/	~86K followers	MS4 premed-mentor creator; relatable study-journey audience of premeds and med students cramming graded material — on-persona urgency.	Micro-influencer collab (authentic study-routine Reel). Ads Manager: lookalike off engagers + interest 'Medical school'. Strong organic-seeding fit.	exam-prep
14	revisign instagram.com/revisign/	~85K followers	Flagship medical-student studygram (Toronto) producing study notes/aesthetics — the deadline-driven note-maker persona Ritsu automates.	Studygram creator-collab target. Whitelist a 'your notes, auto-quizzed' Reel. Ads Manager: lookalike from engagers + interest 'Study skills'. Seed with #studygram.	study-method
15	Codebasics Official instagram.com/codebasicshub/	~59K followers (1M+ learner community)	Data/AI upskilling brand with a large learner community working through structured ML/data courses with deadlines — direct ML-course fit.	Brand-partnership target with a big off-platform community to retarget. Ads Manager: interest 'Data science' + 'Machine learning'; lookalike from engagers.	ml-ai
16	Med School Insiders instagram.com/medschoolinsiders/	~55K followers	Premed/med-student performance-and-study brand (founded by a surgeon) — audience explicitly optimizes study efficiency for graded exams.	Brand-collab + interest target ('Medical school', 'USMLE'). Whitelist an efficiency-angle Reel ('study smarter, not longer'). Lookalike off engagers.	exam-prep

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
17	Blueprint MCAT instagram.com/blueprintmcatprep/	~39K followers	MCAT prep-course brand; followers are mid-prep, deadline-locked science masterers — textbook willingness-to-pay timer for Ritsu.	Competitor-adjacent interest target ('MCAT', 'Test preparation'). Run a 'free practice from your own materials' counter-offer Reel; lookalike from engagers.	exam-prep
18	The Organic Chemistry Tutor instagram.com/theorganicchemistrytutor/	~21K IG followers (huge cross-platform/YouTube)	The go-to brand for STEM-course strugglers (orgo, calculus, physics) — directly names the hardest graded courses Ritsu helps students pass.	Exists as a Meta interest ('The Organic Chemistry Tutor') — strong STEM-course interest lever. Pair with 'Organic chemistry' + 'Calculus' interests for course-specific ad sets.	stem-course
19	Tina Huang (hellotinah) instagram.com/hellotinah/	~20K IG (900K+ YouTube)	Ex-Meta data scientist teaching self-study + AI tech-stack workflows — her self-learner audience overlaps Ritsu's ML-masterer wedge precisely.	Creator-partnership target (she runs a self-learner bootcamp = warm audience). Ads Manager: lookalike from engagers + interest 'Data science' / 'Self-help'.	ml-ai
20	sprinkle studies instagram.com/sprinklestudies/	~50K–100K followers (US nursing studygram)	American nursing-student studygram documenting a graded, deadline-heavy program — US-led, on-persona serious learner converting material to practice.	US studygram micro-collab (high authenticity). Whitelist a 'NCLEX/nursing notes → quiz' Reel. Ads Manager: interest 'Nursing' + 'Study skills'; lookalike off engagers.	study-method
21	#premed instagram.com/explore/tags/premed/	~3M+ posts	Identity tag for pre-professional STEM students facing MCAT/coursework deadlines — concentrated, high-WTP segment of the wedge.	Organic seeding + Reels placement against premed content. For paid, mirror with interest 'Pre-medical' + 'MCAT' and retarget tag-engagers via custom audience.	exam-prep
22	#medschool instagram.com/explore/tags/medschool/	~25M+ posts	One of the largest serious-learner tags; med students are the archetypal dense-material masterers with constant graded deadlines.	Huge organic-reach tag for Reels seeding. Paid mirror: interest 'Medical school' + 'USMLE'; build lookalikes from medical-content engagement audiences.	general-learning
23	#machinelearning instagram.com/explore/tags/machinelearning/	~5M+ posts	Core ML tag where learners working through ML courses/papers congregate — the AI half of the STEM/ML wedge.	Seed ML demo Reels here. Paid mirror: interest 'Machine learning' + 'Deep learning'; US geo-filter for first-100 wedge; retarget tag-engagers.	ml-ai
24	#datascience instagram.com/explore/tags/datascience/	~6M+ posts	Where bootcamp/cohort data-science learners share progress — deadline-bearing ML/DS upskillers Ritsu targets.	Organic seeding tag. Paid mirror: interest 'Data science' + 'Python (programming language)'; lookalike from DS-content engagers.	ml-ai
25	#studywithme instagram.com/explore/tags/studywithme/	~8M+ posts	Tag for focused, session-based studiers (the active-study ritual) — behaviorally the closest to Ritsu's 'do the work' mode.	Strong Reels-seeding tag for 'study with me + Ritsu quiz' content. Paid mirror: interest 'Study skills' + 'Concentration'; retarget engagers.	study-method
26	#studytips instagram.com/explore/tags/studytips/	~5M+ posts	High-intent tag where learners actively seek better study methods — the exact moment to introduce active-recall-as-a-product.	Seed how-to Reels (active recall, Feynman) ending in Ritsu CTA. Paid mirror: interest 'Study skills'; lookalike from study-tip engagement audience.	study-method
27	#gradschool instagram.com/explore/tags/gradschool/	~6M+ posts	Grad students (incl. CS/ML/qualifying-exam) reading dense papers under deadlines — a precise self-learner/researcher slice of the wedge.	Seed 'turn a research paper into a quiz' Reels. Paid mirror: interest 'Graduate school' + 'Research'; retarget tag-engagers; US geo-skew.	general-learning

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
28	#organicchemistry instagram.com/explore/tags/organicchemistry/	~1M+ posts	Names the single most-feared graded STEM course — students here have urgent, deadline-driven mastery needs Ritsu directly serves.	Course-specific Reels seeding ('orgo mechanisms → flashcards'). Paid mirror: interest 'Organic chemistry' + 'Chemistry'; retarget engagers near exam season.	stem-course
29	#calculus instagram.com/explore/tags/calculus/	~1M+ posts	Gateway STEM course for engineering/CS/physics students — broad, deadline-bearing mastery audience for a 'textbook chapter → quiz' demo.	Course-specific seeding tag. Paid mirror: interest 'Calculus' + 'Mathematics'; pair with 'AP Calculus' for exam-season ad sets; retarget engagers.	stem-course
30	#artificialintelligence instagram.com/explore/tags/artificialintelligence/	~10M+ posts	Broadest AI tag capturing learners studying AI/ML who need to convert courses and papers into mastery — wide top-of-funnel for the ML wedge.	Large but broad — seed targeted ML-study Reels and rely on interest 'Artificial intelligence' + narrower 'Machine learning' to refine. Lookalike from AI-content engagers; US geo-filter.	ml-ai

TikTok — creators and hashtags (20)

Ad mechanism: TikTok Ads — interest + hashtag + Spark (creator) Ads.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	#studytips tiktok.com/tag/studytips	~7 billion video views (cumulative)	The single largest study-content hashtag on TikTok and the literal home of active-recall / exam-prep videos this persona watches before a deadline.	Use as a Hashtag interest signal in TikTok Ads Manager and target the #studytips content cluster; run In-Feed ads adjacent to this tag with a 'turn your slides into a quiz' hook.	study-method
2	Justin Sung (Dr Justin Sung) tiktok.com/@justinsungofficial	~300K+ TikTok followers (700K+ YouTube)	A medical doctor and learning-science specialist who preaches active learning over passive re-reading — the EXACT belief Ritsu sells, attracting the rigor-seeking learner.	Top Spark Ads partner: his audience is pre-sold on active recall; boost a Spark Ad framing Ritsu as 'the tool that does the active-learning prep for you' and target lookalikes of his engagers.	study-method
3	Gohar Khan (Gohar's Guide) tiktok.com/@goharsguide	~3.8M TikTok followers	MIT-grad study/college creator whose massive audience is exactly serious US students prepping for hard courses and exams.	Creator-targeting + Spark Ad: pursue a paid partnership or boost his study-method content; his 3.8M serious-student following is a prime custom/lookalike seed audience.	study-method
4	#studytok tiktok.com/tag/studytok	Billions of cumulative views (core StudyTok community tag)	The defining community hashtag for the entire student-study subculture on TikTok — where committed learners self-identify.	Primary hashtag-cluster target; layer with #studytok in interest targeting and seed organic content using this tag to ride the community's native distribution.	study-method
5	#learnontiktok tiktok.com/tag/learnontiktok	Tens of billions of views (TikTok's flagship education tag)	TikTok's official education umbrella hashtag — broadest reach for any learning-intent audience including STEM and AI explainers.	Use the 'Education' interest category plus #learnontiktok adjacency for top-of-funnel reach; pair with a STEM-specific creative to filter for the persona.	general-learning
6	Riley Brown tiktok.com/@rileybrown.ai	~636K TikTok followers	One of TikTok's top AI educators ('number one AI Educator'); his followers are actively learning AI tools and concepts — Ritsu's ML-masterer wedge.	Spark Ads + creator lookalike: target engagers of his AI-education content with a 'master your ML course from the lecture PDFs' angle.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
7	#machinelearning tiktok.com/tag/machinelearning	Hundreds of millions of views	Direct topical hashtag for the ML-course masterer (cs231n/CS229/DeepLearning.AI cohort) who needs to drill dense ML material.	Hashtag + interest targeting on #machinelearning and 'Science & Education' / 'Technology' interests; creative should show a research paper or lecture slide turning into a quiz.	ml-ai
8	Kallaway (Kane Kallaway) tiktok.com/@kanekallaway	~410K TikTok followers	Leading AI/tech explainer creator whose audience is learning-to-use-AI early adopters — high overlap with the AI-curious serious learner.	Spark Ad partnership; target his follower lookalikes with the accuracy/trust angle ('an AI tutor that cites your own source, not hallucinations').	ml-ai
9	Alexis Loveraz tiktok.com/@alexisloveraz	~650K+ followers	Viral STEM tutor teaching algebra, geometry, chemistry, and physics — directly reaches the calculus/chem/physics course-taker segment.	Creator-adjacency + Spark Ad on his STEM tutorial videos; target his audience with subject-specific creatives ('drop your chem chapter, get a quiz in 30s').	stem-course
10	#datascience tiktok.com/tag/datascience	Hundreds of millions of views	Core hashtag for data-science learners and bootcamp cohorts (DataCamp, Zero To Mastery) who must master dense quantitative material on deadline.	Layer #datascience + 'Technology'/'Education' interests; creative angle = converting bootcamp lecture notes and notebooks into active practice.	ml-ai
11	Tina Huang tiktok.com/@hellotinah	~1M across platforms (large TikTok presence; founder of Lonely Octopus)	Ex-Meta data scientist turned educator focused on self-study and data-science skill-building — speaks directly to the self-driven ML/DS masterer.	Spark Ad + lookalike of her self-learner audience; pursue creator collab framing Ritsu as the active-study layer on top of any DS course.	ml-ai
12	#premed tiktok.com/tag/premed	Hundreds of millions of views	Hashtag home of the premed/MCAT crowd — among the most deadline-driven, accuracy-obsessed exam-preppers (the sharpest willingness-to-pay timer).	Target #premed + #mcats hashtag clusters; creative = 'turn First Aid / lecture PDFs into MCAT practice'; this audience pays for any edge before test day.	exam-prep
13	MCATAdvice tiktok.com/@mcataidvice	~67K followers	Dedicated MCAT-prep creator whose entire audience is mid-prep premeds needing active recall over passive content review.	Niche Spark Ad partner with extremely on-persona followers; target engagers with an MCAT-specific 'practice from your own materials' offer.	exam-prep
14	rajistics (Raj) tiktok.com/@rajistics	~31K followers	Explains data science and AI/ML concepts clearly — small but precisely-on-persona audience of technical learners who value accuracy.	Micro-creator Spark Ad with high-intent ML audience; ideal for the trust/accuracy creative ('grounded in your source, not made up').	ml-ai
15	#studywithme tiktok.com/tag/studywithme	Billions of cumulative views	Huge real-time-study-session subculture of students grinding through coursework — captive, focused-study audience.	Run In-Feed ads against #studywithme content; the audience is literally in study mode — strong context for a 'make practice from your notes' CTA.	study-method
16	Deidre Kelly (mathematics teacher) tiktok.com/ @iamthatmathteacher	~1.5M followers	Massive math-focused following (tips and tricks across math topics) covering the calculus/linear-algebra course-taker base.	Creator lookalike + adjacency targeting; her 1.5M math audience is a large seed for math-course creatives ('drop your problem set, get drilled').	stem-course
17	#gradschool tiktok.com/tag/gradschool	Hundreds of millions of views	Hashtag for grad students and qualifying-exam takers breaking down research papers and dense material — a core Ritsu use case.	Target #gradschool + #phdlife clusters with a 'break down a research paper into testable knowledge' creative; high deadline pressure = high intent.	exam-prep

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
18	#examprep tiktok.com/tag/examprep	Tens of millions of views	The literal deadline-trigger hashtag — students searching it are days from a graded exam, exactly Ritsu's 'ace your exam in 3 days' wedge.	High-intent hashtag target; pair with countdown/urgency creative ('exam in 3 days? turn your notes into a quiz now') for bottom-funnel conversion.	exam-prep
19	StudyQuill (Jasmine) tiktok.com/@studyquill	~200K+ followers (large StudyTube/StudyTok crossover)	Well-known study-method creator covering note-taking and effective studying for serious students — strong StudyTok credibility.	Spark Ad partner with a trusted study-method audience; boost content pairing her note systems with Ritsu's auto-generated active practice.	study-method
20	#unistudent tiktok.com/tag/unistudent	~77M+ views	University-student lifestyle/study hashtag reaching the broad college STEM/pre-professional base that forms Ritsu's first-100 wedge.	Mid-funnel hashtag for college-age targeting; combine with 18-24 age + 'Education' interest and a STEM creative to filter to the persona.	general-learning

Discord — servers (20)

Ad mechanism: Organic seeding (Collison-install) — not paid placements.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	Study Together discord.com/invite/study	~1,050,000 members (largest study/productivity server on Discord)	The single biggest concentration of committed self-studiers who run live camera/pomodoro study sessions — the exact passive-to-active-study crowd Ritsu converts.	ORGANIC Collison-install seeding (no Discord self-serve ad targeting of servers). Partner with admins/mods for a sponsored 'study tool' channel or pinned resource; run a free-Ritsu giveaway tied to an exam push; seed share-links in resource/tool-recommendation channels where rules allow.	general-learning
2	Mathematics discord.com/invite/math	~332,000 members	Huge population learning linear algebra, calculus, and proof-based math — the STEM foundation courses with hard graded deadlines that Ritsu turns into quizzes/flashcards.	ORGANIC seeding. Be genuinely helpful in homework/help channels, then demo Ritsu turning a lecture PDF into a quiz; pursue a mod partnership for a study-resources pin. No paid placement available — earn trust first.	stem-course
3	Hugging Face discord.com/invite/hugging-face-879548962464493619	~221,000 members	Largest hands-on ML/AI practitioner hub (papers, courses, NLP/agents tracks) — learners working through dense ML material who value accuracy and rigor.	ORGANIC seeding via the learning/course channels (agents-course, deep-RL-course cohorts). Position Ritsu as a paper-and-lecture-to-active-practice tool; an AI-literate audience that will judge accuracy harshly, so lead with grounded/cited-quiz messaging.	ml-ai
4	The Coding Den discord.com/invite/code	~176,000 members	Largest general programming-help community; engineers and CS students learning new languages/frameworks from docs — Ritsu's 'learn a framework fast' workflow.	ORGANIC seeding in help/resources channels. Demo Ritsu turning framework docs or a CS course PDF into code exercises + flashcards; mod relationship needed before any link-drop (strict anti-spam culture).	stem-course
5	The Programmer's Hangout discord.com/invite/programming	~170,000+ members	Massive CS/dev community spanning beginners to seniors learning DS&A, web, mobile — the engineer archetype mastering technical material on a deadline.	ORGANIC seeding. Contribute in #resources and study-group channels; partner with staff for a curated-tools mention. Treat as awareness/seeding, not direct-response — no native ad units.	stem-course

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
6	Learn AI Together (Towards AI) discord.gg/learnaitogether	~80,000+ members	Largest AI-learning-focused community (weekly reading group, course discussion, job board) — exactly the 'committed ML masterer working through courses/papers' persona.	ORGANIC seeding; highest persona precision on the platform. Sponsor or contribute to the weekly paper-reading group; offer Ritsu free codes to members prepping for ML coursework. Coordinate with Towards AI team for a partnership/AMA.	ml-ai
7	Science and Technology discord.com/invite/science	~57,000 members	Broad STEM community (physics, chem, CS, bio) where students bring coursework and competition prep — multi-subject dated-deadline learners.	ORGANIC seeding across subject help channels. Lead with accuracy/no-hallucinated-quizzes messaging (science crowd is skeptical of AI); seek a mod-pinned resource. No paid targeting available.	stem-course
8	AP Students discord.com/invite/apstudents	~55,000 members	Affiliated with r/APStudents; students with a hard, DATED College Board exam (AP Calc, Physics, Chem, CS-A) — the sharpest 'ace the exam in N days' willingness-to-pay timer.	ORGANIC seeding, peaks Mar–May (AP season). Offer free Ritsu during the exam crunch; partner with mods for a study-resource pin. Tightest deadline-driven wedge — message the 3-day-cram job-to-be-done directly.	exam-prep
9	MCAT / Premed discord.com/invite/premed	~30,000 members	Pre-med students grinding the MCAT — a high-stakes, dated, content-dense exam where active recall on dense material is the core job Ritsu serves.	ORGANIC seeding in content-review and AAMC-FL channels. Demo turning dense bio/biochem/physics review PDFs into Anki-style decks + quizzes; high willingness-to-pay (already buy UWorld/Anki). Partner with mods for a tools pin.	exam-prep
10	fast.ai (official Discord) forums.fast.ai/t/fast-ai-discord/122405	Tens of thousands (official server for the fast.ai 'Practical Deep Learning' course community)	Cohort of learners actively working through the fast.ai course — the canonical 'committed ML masterer on a multi-week course' archetype.	ORGANIC seeding only; small, tight, high-trust community — relationship-first. Help in study channels and offer Ritsu to turn lecture videos/notebooks into recall practice. No link-spam; earn an admin mention. Invite link is gated via the fast.ai forums thread.	ml-ai
11	DeepLearning.AI (Andrew Ng community) community.deeplearning.ai/ (Discord groups linked per-specialization)	Large (per-specialization Discord groups + a 100k+ member forum community)	Learners enrolled in Andrew Ng / Coursera ML specializations — paying, deadline-bearing course-takers who are the definitional Ritsu wedge.	ORGANIC seeding within specialization Discord groups (each course spins its own). Coordinate with course community managers; offer Ritsu to convert lecture slides into quizzes. Primarily a forum-linked Discord, so seed via the official community hub.	ml-ai
12	Biggest Physics disboard.org/servers/tag/physics (top-ranked 'Biggest Physics' server)	~20,000–40,000 members (largest physics-specific help server)	Physics students prepping school/college coursework and exams (JEE, NEET, A-levels, AP, SAT) — dated-deadline STEM-course masterers.	ORGANIC seeding in homework/exam-prep channels. Demo turning a physics chapter or lecture into problem sets; lead with accuracy (physics answers are verifiable). Mod partnership for a resource pin; no paid units.	stem-course
13	StudyStream discord.gg/studystream	~100,000+ members (large live-study community; flagship is the web app + Discord)	Committed students running live focus/study-room sessions worldwide — passive-study-burned learners who would adopt an active-recall tool.	ORGANIC seeding plus potential brand partnership (StudyStream runs sponsorships with study brands). Pitch a co-branded study challenge or resource integration; seed share-links in tool/resource channels with admin buy-in.	general-learning

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
14	MLOps Community discord.gg/mlopscommunity	~20,000+ members (Discord/Slack; community founded by Demetrios Brinkmann)	ML engineers and data scientists upskilling via courses, papers, and certs — the professional end of the committed-ML-masterer persona.	ORGANIC seeding via learning/career channels; sponsor or join a reading-group/event. Position Ritsu for cert/course prep (e.g., cloud-ML certs). Practitioner audience — credibility and accuracy framing essential.	ml-ai
15	Data Science (Chip Huyen community) discord.gg/datascience (community started by Chip Huyen)	~20,000+ members	Data scientists and ML learners with regular events, career advice, and study discussion — deadline-bearing learners working through DS/ML curricula.	ORGANIC seeding in learning-resources channels. Demo Ritsu converting DS course material / textbook chapters into quizzes and concept maps; coordinate with organizers for an events/AMA slot. No native ad targeting.	ml-ai
16	STEM Students (Science, Technology, Engineering & Maths) discord.do/stem-students-science-technology-engineering-maths/	~15,000–30,000 members	Cross-disciplinary STEM help community (math, physics, chem, bio, CS, programming) where students bring graded coursework — broad multi-subject wedge.	ORGANIC seeding across the per-subject help channels. Be the helpful expert, then show Ritsu auto-generating practice from a chapter; mod-pinned tool mention. Seeding only — no paid placement.	stem-course
17	Optimal Learning discord.me/optimallearning	~5,000–15,000 members	Free tutoring server for math (algebra→linear algebra, diff eq) and science (bio, chem, CS, physics) — students actively seeking help to master coursework.	ORGANIC seeding; small enough that a tutor/mod partnership is achievable fast. Offer Ritsu as a self-serve practice layer between tutoring sessions; high intent-to-learn audience. No ad units — relationship-driven.	study-method
18	Kaggle (community-run ML Discord) discord.gg/kaggle (community server; channels for competitions, notebooks, learning)	~15,000+ members (community-run, not official Kaggle)	Competitive ML learners sharing notebooks and learning resources — self-driven masterers working through datasets and ML technique on competition deadlines.	ORGANIC seeding in learning-resources/notebook channels. Frame Ritsu for turning ML course content and competition write-ups into recall practice; deadline framing works (competition end dates). Earn mod trust before linking.	ml-ai
19	STEM Solutions discordhome.com/server/stemsolutions	~5,000–10,000 members	Tutoring-focused STEM server (math, CS, engineering, sciences) staffed by tutors and professors — students paying attention specifically to mastering hard graded material.	ORGANIC seeding plus a possible tutor-partnership (tutors could recommend Ritsu for between-session practice). High purchase intent (already engaging paid-ish help). No native targeting — partner with the tutor team.	study-method
20	Clarity Learning discordhome.com/ (search 'Clarity Learning'; ~1,400+ member coursework-help server)	~1,400+ members	Small, high-intent server dedicated to helping students excel in coursework — concentrated deadline-bearing learners ideal for a tight Collison-install pilot.	ORGANIC seeding; ideal first-pilot server (small enough to onboard members by name per SOP-CUSTOMER-006). Founder-led personal onboarding + free Ritsu codes; measure activation before scaling to larger servers. No paid ads.	study-method

LinkedIn — groups and influencers (18)

Ad mechanism: LinkedIn Ads — group + skill + job-title(Student) + field-of-study.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	Andrew Ng linkedin.com/in/andrewyng/	~2.35M followers	Founder of DeepLearning.AI / Coursera ML courses and fast.ai-adjacent education — the single most likely person whose graded course this persona is actively studying for a deadline.	Cannot target his followers directly on LinkedIn Ads; seed organically (comment/native posts), build an engagement Matched Audience from any co-posted content, and pair with interest-targeting on 'DeepLearning.AI' / 'Coursera' + member-skill 'Machine Learning'.	ml-ai
2	Artificial Intelligence, Machine Learning, Data Science & Robotics (LinkedIn Group) linkedin.com/groups/3990648	~3.0M members	The largest AI/ML community on LinkedIn — maximum reach into the broad ML-learner pool the persona sits inside.	LinkedIn Ads supports Group targeting: add this group, then layer member-skill 'Machine Learning' + 'Deep Learning' and Job Title = Student / field-of-study = Computer Science to sharpen toward enrolled learners; also viable for organic seeding posts.	ml-ai
3	Sebastian Raschka, PhD linkedin.com/in/sebastianraschka/	~194K followers (77K 'Ahead of AI' newsletter subs)	Author of widely-used ML/LLM textbooks and a Wisconsin ML professor — exactly the rigorous, accuracy-first voice this skeptical 'will-it-lie-to-me' masterer trusts when studying from his books.	Seed organically and use a follower-lookalike via member-skills ('Machine Learning', 'PyTorch', 'Deep Learning') + interest 'Build a Large Language Model From Scratch'; retarget engagers of any Ritsu post that references his material.	ml-ai
4	Artificial Intelligence, Deep Learning, Machine Learning (LinkedIn Group) linkedin.com/groups/1814785/	~2.55M members	Second-largest technical AI/DL group, skewed toward deep-learning research and coursework — strong fit for the cs231n / 6.S191 / DeepLearning.AI cohort.	Add as a Group facet in LinkedIn Campaign Manager; intersect with member-skill 'Deep Learning' + 'Neural Networks' and degree/field-of-study filters to bias toward currently-enrolled DL students.	ml-ai
5	Andriy Burkov ca.linkedin.com/in/andriyburkov	~1.0M followers	Author of 'The Hundred-Page Machine Learning Book' / 'The Hundred-Page Language Models Book' — a canonical exam-prep text this persona reads to master ML fast under deadline.	Organic-seed + build interest/skill lookalike ('Machine Learning' skill, interest in his books); retarget content engagers. Direct follower targeting is not available, so use his book titles as interest signals.	ml-ai
6	Machine Learning Community (Moderated) (LinkedIn Group) linkedin.com/groups/961087/	~2.0M members	Large moderated ML community (higher signal-to-noise than open groups), good for reaching serious practitioners and students who self-curate.	Use Group targeting + tighten with member-skill 'Machine Learning' and Job Title contains 'Student' / 'Research Assistant'; moderated nature also makes it a candidate for a well-crafted organic seed post.	ml-ai
7	Barbara Oakley, PhD (Learning How to Learn) linkedin.com/in/barbaraoakley/	~50K+ followers (course: 4M+ learners on Coursera)	Creator of 'Learning How to Learn' — owns the exact learning-science angle (active beats passive recall) that resonates with a learner burned by re-reading and re-watching.	Highest message-fit for Ritsu's 'active mastery beats passive study' hook; seed organically and build an interest audience around 'Learning How to Learn' / spaced-repetition / retrieval-practice plus Job Title = Student to reach study-method-minded learners.	study-method
8	Ai Mindshare - AI, Machine Learning, ML, Data Science (LinkedIn Group) linkedin.com/groups/60878/	~1.23M members	Active AI/ML + GenAI/LLM community — captures learners studying current LLM coursework and bootcamp content.	Group facet + member-skill 'Generative AI' / 'Large Language Models'; combine with field-of-study = Data Science / Computer Science to bias toward enrolled learners over working pros.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
9	Data Science Central (LinkedIn Group) linkedin.com/groups/35222/	~330K+ members	Long-running data-science learning hub (articles, webinars, forums) where DS bootcamp and course students gather.	Target the group + member-skills 'Data Science', 'Python', 'Statistics'; overlay Job Title = Student and seniority = Entry/Training to reach bootcamp cohorts (Springboard / DataCamp / ZTM type learners).	ml-ai
10	Cassie Kozyrkov linkedin.com/in/kozyrkov/	~700K+ followers	Former Google Chief Decision Scientist and prolific explainer of statistics/ML fundamentals — the rigor-and-intuition voice this accuracy-driven masterer follows to actually understand concepts.	Seed organically; build a skill/interest lookalike ('Statistics', 'Machine Learning', 'Decision Intelligence'). Her audience over-indexes on 'understand it properly' which matches Ritsu's anti-cheat mastery framing.	ml-ai
11	Learning, Education, and Training Professionals Group (LinkedIn Group) linkedin.com/groups/27003/	~1.35M members	Largest learning/education community — reaches educators (one of Ritsu's six personas) plus serious self-learners interested in study methods.	Group targeting for the educator + lifelong-learner angle; layer interest 'instructional design' / 'study skills'. Best for the 'create a course / master a subject' message rather than the ML-specific one.	general-learning
12	Brain Expansion Group (LinkedIn Group) linkedin.com/groups/13927028	~2.54M members	Huge learning / personal-growth / emerging-tech community — broad top-of-funnel reach for the 'master tough subjects' message.	Use for scale on the general-learning hook; tighten hard with member-skill + Job Title = Student to avoid generic self-help audiences. Treat as a reach amplifier, not a precision facet.	general-learning
13	KDnuggets — AI, Big Data, Data Science, Machine Learning (LinkedIn Group + Company) linkedin.com/company/kdnuggets	Group ~150K members; company page audience additional	Authoritative DS/ML learning brand whose group is full of people actively studying tutorials and prepping technical skills.	Target the moderated group + KDnuggets company-page followers via interest; member-skill 'Data Mining'/'Machine Learning'. Strong candidate for a sponsored-content placement aimed at active learners.	ml-ai
14	Ken Jee linkedin.com/in/kenjee/	~200K+ followers (#66DaysOfData community)	Runs the #66DaysOfData challenge and data-science learning content — directly mobilizes deadline/streak-driven self-learners, mirroring Ritsu's streak and active-practice loop.	Seed organically around #66DaysOfData; build engagement/interest audiences from that hashtag + member-skill 'Data Science'. His streak-challenge framing aligns with Ritsu's emotional-hook + heatmap.	study-method
15	Daniel Bourke (mrdbourke) linkedin.com/in/mrdbourke/	~60K+ followers (large YouTube ML-course audience)	Creator of popular 'learn ML / PyTorch from scratch' courses — his audience is self-learners mid-way through a hard ML curriculum, exactly Ritsu's wedge.	Organic-seed + interest lookalike on 'PyTorch', 'Machine Learning', 'TensorFlow' and field-of-study = Computer Science; retarget engagers of any PyTorch-course-themed Ritsu post.	ml-ai
16	Steve Nouri linkedin.com/in/stevenouri/	~1.6M followers	One of LinkedIn's largest AI Top Voices; his feed is a daily on-ramp for people learning AI/GenAI, providing broad reach into the ML-curious student pool.	Reach play, not precision: seed organically and build a broad interest audience ('Artificial Intelligence', 'Generative AI'); intersect with Job Title = Student to filter toward learners vs. executives.	ml-ai
17	Python Data Science, Machine Learning, Graph, and NLP (LinkedIn Group) linkedin.com/groups/8458811/	~92K members	Niche, Python-first DS/ML/NLP group — precisely the technical-coursework audience (Python + ML + NLP) for the STEM/CS-course slice of the persona.	High-precision group facet; combine with member-skills 'Python', 'Natural Language Processing', 'Machine Learning'. Smaller but on-persona — ideal for a tightly-scoped sponsored test.	stem-course

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
18	Lex Fridman linkedin.com/in/lexfridman/	~1.48M followers	MIT researcher and podcaster whose AI/deep-learning audience includes large numbers of students and self-learners aspiring to master the field.	Top-of-funnel reach via organic seeding + a broad AI/robotics interest audience; tighten with member-skill 'Deep Learning' + field-of-study = Computer Science / EECS to bias toward enrolled learners rather than general podcast fans.	ml-ai

Podcasts — shows (18)

Ad mechanism: Host-read sponsorship slots.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	Huberman Lab hubermanlab.com/podcast	~5.4M+ subscribers; millions of downloads per episode; routinely top-10 global, #1 in Science/Education	The single biggest hub for the persona's exact mindset — Stanford neuroscientist Andrew Huberman has dedicated episodes ('Optimal Protocols for Studying & Learning', 'How to Learn Skills Faster') on test-based active recall, spaced repetition, and gap effects, which is literally Ritsu's product thesis.	Buy host-read mid-roll sponsorship slots specifically on the learning/focus/memory episodes (evergreen, replayed for years); pitch Ritsu as 'the active-recall tool Huberman's science says actually works.' Target via podcast ad networks supporting episode-level placement (e.g., the show's ad sales / dynamic-insertion provider).	study-method
2	Lex Fridman Podcast lexfridman.com/podcast/ (Apple: id1434243584)	~3.6M+ subscribers; multi-million plays per episode across YouTube/Spotify/Apple	Massive AI/ML-curious technical audience (MIT background host; episodes with Terence Tao, Andrew Ng, Edward Frenkel on math/physics/AI) that maps directly to the ML/STEM masterer who wants rigor, not chatbot fluff.	Host-read sponsorship is premium and high-trust; if full-show is out of budget, use dynamic ad insertion targeted at the AI/math/physics episodes. Angle: 'turn these dense lectures and papers into active mastery.' Skeptical-of-AI audience responds to the accuracy/citations message.	ml-ai
3	The TWIML AI Podcast (This Week in Machine Learning & AI) twimlai.com (Apple: id1116303051)	7M+ downloads lifetime; large, sustained ML-practitioner community	The most on-persona large ML show — serious practitioners and students learning ML deeply from research interviews, exactly the cohort taking cs231n/CS229/DeepLearning.AI who need to convert papers into practice.	Host-read sponsorship with Sam Charrington carries strong credibility with ML learners; pitch the 'research-paper → quiz/flashcards/concept-map' workflow. Bundle with the show's newsletter for cross-channel reach.	ml-ai
4	Latent Space: The AI Engineer Podcast latent.space/podcast (Apple: id1674008350)	200K+ subs / 10M+ readers+listeners across channels in 2025	The reference show for the engineer-learning-ML archetype (fast.ai /DeepLearning.AI grads building with LLMs) — precisely the 'learning a new framework from docs' persona who'd use Ritsu to master dense technical material fast.	Newsletter+podcast combo from swyx/Alessio; host-read or newsletter-sponsor slots reach AI engineers directly. Angle: 'master a new framework or paper in 30 seconds of upload.' Strong fit for the engineer sub-persona's WTP.	ml-ai
5	Super Data Science: ML & AI Podcast superdatascience.com/podcast (Apple: id1163599059)	987+ episodes, semiweekly (Tue/Fri); one of the largest data-science learning shows	Explicitly a learning-and-career show for data science / ML students and bootcampers (host Jon Krohn teaches), matching the paid-bootcamp-cohort archetype with a dated curriculum and graded outcomes.	High episode cadence = frequent, affordable host-read slots; audience is actively upskilling (perfect intent). Pitch Ritsu as the study layer on top of their DS/ML coursework. Two episodes/week means cheap A/B testing of ad creatives.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
6	Deep Questions with Cal Newport podcasts.apple.com/us/podcast/deep-questions-with-cal-newport/id1515786216	100K+ newsletter subscribers; large weekly podcast audience	Hosted by a CS professor; the definitive show on deep work, focus, and student productivity — its audience self-selects as serious, deadline-driven learners frustrated by shallow study, the core Ritsu buyer.	Host-read ads land well because Newport vets sponsors and his audience trusts study/productivity tools. Angle: 'deep work needs active practice, not re-reading — Ritsu builds the practice.' Avoid hypey creative; this audience prizes honest/concrete (matches Ritsu brand voice).	study-method
7	Divine Intervention Podcasts (USMLE) divineinterventionpodcasts.com/ (Apple: id1483304964)	Very large within US med-student niche; one of the most-listened USMLE Step 1-3 / shelf prep shows	Pure exam-prep persona with a hard dated licensing exam — med students with the sharpest 'pass-this-graded-deadline' willingness-to-pay, exactly the wedge Ritsu's 'ace your exam' workflow targets.	Niche but extremely high-intent; host-read or pre-roll reaches students mid-grind. Pitch: 'upload your First Aid / lecture PDFs → instant high-yield quizzes.' Cost-efficient because audience is concentrated and purchase-ready before Step exams.	exam-prep
8	MCAT Basics (from MedSchoolCoach) podcasts.apple.com/us/podcast/mcat-basics-from-medschoolcoach/id1448973714	Leading MCAT-prep podcast; large pre-med listener base in the US	Pre-med students prepping a dated, high-stakes standardized exam — the 'committed, deadline-bearing' persona at peak study intensity, currently leaning on Anki/Quizlet that Ritsu replaces.	Host-read endorsement from a known prep brand transfers trust; target as a sponsor of section-review episodes. Angle: 'turn AAMC content and your prep books into adaptive practice.' Seasonal spend around MCAT test dates maximizes ROI.	exam-prep
9	Machine Learning Street Talk (MLST) podcasts.apple.com/us/podcast/machine-learning-street-talk (rephonic: machine-learning-street-talk)	249+ episodes; respected, sizeable deep-ML/theory audience	Deep, theory-heavy ML discussions for the rigor-seeking learner (the cs231n/CS229 crowd who distrusts surface-level explanations) — an audience that values accuracy, Ritsu's #1 service-quality promise.	Smaller but extremely on-persona and intellectually demanding; host-read works only with an honest, technical pitch. Angle: 'grounded, cited mastery — not a hallucinating chatbot.' Great for credibility-building reads that this skeptical cohort respects.	ml-ai
10	Data Skeptic podcasts.apple.com/ (millionpodcasts: data-science-podcasts) / dataskeptic.com	200+ episodes, weekly; one of the best-known long-running data-science shows	Audience is literally self-selected skeptics in data science/stats/ML — the 'will the AI lie to me?' learner who must be convinced Ritsu is accurate before trusting it with exam prep.	The skeptic framing is an asset: a candid host-read about citations + on-document grounding converts the doubters. Pitch the accuracy/reliability SERVQUAL message head-on. Weekly cadence keeps slot pricing reasonable.	ml-ai
11	The Cognitive Revolution (AI Builders & Researchers) podcasts.apple.com/us/podcast/the-cognitive-revolution/ id1669813431	Biweekly; substantial and fast-growing AI audience (Turpentine network)	Reaches AI-curious learners and builders going deep on how AI works — adjacent to the ML-masterer who wants to genuinely understand the field rather than skim it.	Part of the Turpentine network, so host-read inventory is professionally sold and can be bundled across the network's AI shows. Angle: 'understand AI by actively mastering the papers, not just listening.' Good scaled reach to AI enthusiasts.	ml-ai
12	The Learning Scientists Podcast learningscientists.org/podcast	Niche but authoritative; strong reach among educators and study-method-serious students	Hosted by cognitive psychologists on the exact science Ritsu operationalizes (retrieval practice, spacing, interleaving) — its listeners already believe active learning beats passive, so they're pre-sold on Ritsu's premise.	Small audience, maximum message-fit; host-read endorsement from learning scientists is the highest-credibility validation possible. Pitch: 'we automate retrieval practice and spacing from any document.' Ideal for an authority-building seed placement.	study-method

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
13	Gradient Dissent (Weights & Biases) wandb.ai/fully-connected/gradient-dissent (Apple/Spotify: Gradient Dissent)	Established applied-ML practitioner audience (W&B community)	Applied-ML show whose listeners are practitioners and students learning to actually build models — the engineer/data-science masterer converting docs and courses into working skill.	Host-read via the W&B brand reaches ML engineers with high trust; pitch the 'learn a new ML framework/paper fast' angle. Tight thematic alignment makes a credible, low-waste placement.	ml-ai
14	Practical AI practicalai.fm (Changelog network)	Long-running weekly show; large Changelog-network developer/ML audience	Makes ML 'practical, productive, accessible' for learners moving from fundamentals to LLMs/MLOps — the self-learner and engineer building ML competence who'd use Ritsu to master technical material.	Sold through the Changelog network's sponsorship program (developer-trusted, host-read). Angle: 'from fundamentals to mastery — upload the course, get active practice.' Network buy enables cross-show reach to developers/ML learners.	ml-ai
15	Learn and Teach Better with Dr. Justin Sung (iCanStudy / guest appearances) youtube.com/@JustinSung (40M+ YouTube views; podcast guest + iCanStudy content)	20,000+ trained students in 150+ countries; 40M+ YouTube views on study-method content	Ex-doctor learning coach whose entire audience is students hunting better study methods (encoding, active learning) — the most behaviorally aligned 'how do I actually learn this?' cohort for Ritsu.	Best reached via host-read/sponsored segments on the podcasts he guests on and his own channels; his endorsement of an active-learning tool is gold for the study-method buyer. Pitch the encoding/active-recall automation angle directly.	study-method
16	My Favorite Theorem podcasts.apple.com/us/podcast/my-favorite-theorem/id1262231136	Niche; audience is mostly math-interested undergraduates and math grad students/faculty	Audience is explicitly math undergrads and grad students — the linear-algebra/calculus/proof-heavy STEM-course persona who needs to turn dense theorems and lecture notes into testable practice.	Small but laser-targeted at the math-course learner; host-read works with a numerate, non-hypey pitch. Angle: 'turn your problem sets and lecture PDFs into quizzes and concept maps.' Cheap, precise placement for the math sub-segment.	stem-course
17	The Knowledge Project (Shane Parrish / Farnam Street) fs.blog/knowledge-project/	Large general-learning audience; one of the biggest mental-models/learning shows	Big audience of self-improvement-driven learners obsessed with thinking and mastery — the 'serious self-learner' persona running multi-source learning projects that Ritsu is built for.	Host-read sponsorship reaches a broad, affluent, learning-motivated audience; pitch Ritsu as the tool that turns any source into mastery. Good top-of-funnel scale for the self-learner segment beyond the narrow exam crowd.	general-learning
18	Deep Dive with Ali Abdaal aliabdaal.com/podcast/	Large; built on Ali Abdaal's multi-million-subscriber productivity/study brand (ex-doctor, study-tips creator)	Abdaal's audience came up through his evidence-based study-technique content (active recall, spaced repetition) — students and self-learners optimizing how they study, a direct match for Ritsu's active-mastery pitch.	Host-read endorsement carries heavy weight with the study-optimization audience that already practices active recall; pitch Ritsu as the automation of the techniques he teaches. Strong creator-credibility transfer to the study-method buyer.	study-method

Newsletters / Substack — newsletters (18)

Ad mechanism: Newsletter sponsorship / classified slots.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	Ahead of AI (Sebastian Raschka) magazine.sebastianraschka.com / (Substack @rasbt)	~190K-193K subscribers (ranked #56 in Technology on Substack)	The single most on-persona newsletter: Raschka teaches LLM/ML from-scratch, read by ML students and self-learners working through exactly the cs231n/CS229/fast.ai material Ritsu turns into practice.	Substack-native: pursue a recommendation/cross-promotion or a sponsored classified in the issue body; Raschka curates a 'Substacks recommended' list — getting listed there is a high-trust organic seed to a skeptical, accuracy-valuing audience. Pitch the 'turn this paper/lecture into a quiz' angle.	ml-ai
2	The Batch (DeepLearning.AI / Andrew Ng) deeplearning.ai/the-batch	Hundreds of thousands (DeepLearning.AI's flagship weekly; audience overlaps the millions of Ng/DeepLearning.AI course-takers)	Directly reaches the DeepLearning.AI / Coursera / Andrew Ng course-takers who ARE the deadline-bearing ML-course persona; brand authority neutralizes the 'will AI lie to me?' objection.	Not a self-serve Substack — request the DeepLearning.AI media kit / partnership slot. Premium, high-credibility placement; frame Ritsu as the active-practice companion to their courses (Andrew Ng audience is mastery-motivated, not novelty-seeking).	ml-ai
3	TLDR AI tldr.tech/ai	~1.25M+ subscribers (daily; engineers + technical professionals)	Largest technically-skewed AI list; its engineer/student readers are precisely those learning new frameworks from docs and dense papers — Ritsu's 'learn a framework fast' and 'break down a research paper' jobs.	Self-serve-ish sponsorship via tldr.tech/advertise; classified + primary slots, CPM-priced (\$1K-\$20K+ range). Ask for the AI-vertical-only send to keep spend on-persona. Lead with the technical-accuracy / cited-from-your-source proof point.	ml-ai
4	AlphaSignal alphasignal.ai/ (Lior Sinclair)	~200K+ AI developers/researchers	Research-to-practice newsletter for engineers building AI systems — the skeptical, rigor-first reader who wants to genuinely master papers, not skim them.	Direct sponsorship inquiry via the site; weekly tech-heavy send means one placement reaches a concentrated ML-practitioner base. Emphasize Ritsu's grounded-in-your-source citations (this audience punishes hallucination).	ml-ai
5	Ali Abdaal's Newsletter (LifeNotes / Sunday Snippets) aliabdaal.com/newsletter/ (also Substack: aliabdaal.substack.com)	~250K+ subscribers	The definitive study-skills/evidence-based-learning audience (Abdaal built his brand on active recall + spaced repetition for med/exam study) — the exact psychographic of the serious learner burned by passive study.	Premium creator sponsorship via his team/media kit (single sends reportedly out-earn a month of doctor salary). Highest-intent fit in the list: pitch Ritsu as the tool that automates the active-recall material his audience already believes in.	study-method
6	The Rundown AI therundown.ai/ (Rowan Cheung)	~2M+ subscribers	Largest AI newsletter overall with strong student/early-career-learner readership; broad reach to people actively trying to skill up in AI on a timeline.	Sponsorship via therundown.ai advertise page (high open rate 50%+). Highest raw reach — use for top-of-funnel awareness; request the most AI-education-relevant segment and A/B the 'master, don't just chat' hook against the cheating objection.	ml-ai
7	Superhuman AI (Zain Kahn) superhuman.ai/	~1.5M+ subscribers	Massive learn-to-use-AI audience of professionals and students upskilling; reaches career-changers and self-learners (two of Ritsu's six personas) at scale.	Self-serve sponsorship marketplace (superhuman.ai); classifieds + dedicated slots. Strong for volume; pair the ad with a free-tier 'upload your course PDF, get a quiz in 30s' CTA to convert the curious into activation.	general-learning

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
8	Import AI (Jack Clark) importai.substack.com/ (Substack @importai)	~100K+ subscribers	Deep, research-literate readership (Anthropic co-founder's analysis of papers/policy) — the rigor-obsessed researcher persona who breaks down dense papers into testable knowledge.	Substack-native: sponsored mention in-issue or Substack Recommendations cross-promo. Smaller but extremely high-quality/credible audience; ideal for the 'researcher breaking down a paper' use case. Keep copy substantive — this audience distrusts hype.	ml-ai
9	The Neuron theneurondaily.com/	~700K+ subscribers	Large daily AI newsletter whose readers are actively learning to apply AI; broad learner/professional overlap with Ritsu's self-learner and professional personas.	Sponsorship via The Neuron / TechnologyAdvice (acquired Jan 2025 — now has formal ad sales). Proven paid-acquisition channel themselves (they bought Meta ads to grow), so sponsorship inventory is mature. Good mid-funnel reach.	general-learning
10	Deep Learning Weekly deeplearningweekly.com/	~30K subscribers	Curated research papers + tutorials for ML practitioners — a small but laser-targeted list of people consuming exactly the dense ML material Ritsu converts to active practice.	Direct sponsorship via site; low-cost, high-precision placement. Best efficiency-per-dollar for the core ML-masterer wedge. Lead with 'turn this week's paper into flashcards + a concept map.'	ml-ai
11	The Gradient thegradient.pub/ (also Substack)	Tens of thousands (volunteer-run ML research publication + newsletter/ podcast)	In-depth ML research analysis read by grad students and serious researchers — the audience facing qualifying exams and dense-paper mastery.	Substack-native sponsored placement or podcast read; academically credible context. Niche but precisely on the grad-school/research segment of the persona. Frame around exam/quals prep from papers.	ml-ai
12	Scott H. Young Newsletter (Ultralearning) scotthyoun.com/blog/ newsletter-ss/ (Substack)	Tens of thousands (long-running ultralearning audience; author of 'Ultralearning')	The ultralearning / learn-hard-things-fast audience — self-identified serious learners who deliberately attack hard STEM material on a deadline, exactly Ritsu's psychographic.	Substack sponsorship / Recommendations cross-promo. Smaller list but unusually high mastery-intent; pitch Ritsu as the logistics engine for the spaced-repetition + active practice his readers already practice (he's even written about AI helping with spacing).	study-method
13	Data Elixir dataelixir.com/	Tens of thousands (long-established weekly data-science curation)	Curated data-science/ML articles, tools, and tutorials for practitioners and students learning the field — overlaps the data-science-bootcamp segment of the persona.	Self-serve sponsor slots via dataelixir.com (classified + featured). Reliable, on-topic data-science reach at modest cost; good for the DataCamp/bootcamp-cohort variant of the wedge.	ml-ai
14	Data Science Weekly datascienceweekly.substack.com /	Tens of thousands of subscribers	Long-running curated DS newsletter read by learners and career-changers entering data science — Ritsu's 'multi-source learning project' and bootcamp learners.	Substack-native sponsored placement; affordable and precisely on the data-science learning niche. Pair with the 'turn course slides + textbook into one study plan' CTA.	ml-ai
15	KDnuggets Newsletter kdnuggets.com/news/ subscribe.html	~60K+ subscribers	Veteran data-science/ML education hub; audience includes students, course-takers, and practitioners studying ML fundamentals on a timeline.	Direct ad/sponsorship sales via KDnuggets (display + newsletter slots); established education-focused publisher. Good for credible reach into the broad DS/ML-learner base.	ml-ai

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
16	Justin Sung / iCanStudy Newsletter (Learning Drops) icanstudy.com/ (newsletter signup; Justin Sung)	Tens of thousands (learning-science newsletter tied to a large YouTube/ learning-method following)	Pure learning-science audience: Sung teaches higher-order learning, mind-mapping, and spaced-interleaved-retrieval to students prepping hard exams — the persona's exact study philosophy.	Direct sponsorship / partnership inquiry (he openly evaluates EdTech tools). High strategic fit — a credibility co-sign from a learning-science authority strongly counters the 'AI can't teach real mastery' objection. Pitch Ritsu's concept-map + Feynman/ active-recall modes.	study-method
17	Mindstream mindstream.news/	~150K-200K+ subscribers (acquired by HubSpot)	Large daily AI newsletter with a learn-to-use-AI bent; broad reach to upskilling students and professionals for top-of-funnel awareness.	Sponsorship via Mindstream/HubSpot ad sales (mature inventory post-acquisition). Use for scaled awareness; gate to the most education-relevant segment and lead with the free-tier activation hook.	general-learning
18	Last Week in AI lastweekin.ai/ (newsletter + podcast)	~30K subscribers	Research-recap newsletter/podcast for AI-literate learners; small but engaged audience of people actively tracking and studying ML developments.	Direct sponsorship (newsletter slot + podcast read). Niche, high-engagement; cost-efficient supplement to the larger placements. Frame around staying current AND mastering the underlying material via active practice.	ml-ai

Forums & communities — forums and communities (18)

Ad mechanism: Organic seeding + community participation.

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
1	Kaggle (Discussions + Competitions forums) kaggle.com/discussions	~31M registered members (largest data-science/ML community; expected ~30M registered by Q2 2026)	The single largest gathering of ML/data-science learners working through dated competitions and graded coursework who need to turn papers/notebooks/lectures into mastery fast.	No native self-serve ad manager — ORGANIC seeding only: post a high-value notebook/dataset (e.g. 'turn the cs231n PDF into a quiz deck'), answer in Getting Started + course-competition discussion threads, and sponsor/seed via Kaggle Grandmaster creators. Treat as community-partnership, not paid CPC.	ml-ai
2	Hugging Face Forums discuss.huggingface.co/	~5M HF platform users; Discourse forum has tens of thousands of active threads (Beginners, Course, Research categories)	Home of self-taught ML masterers working through the HF course and deep-learning material — exactly the 'turn dense ML content into active practice' job.	Discourse forum, no ad system — ORGANIC: be genuinely useful in the 'Beginners' and 'Course' categories, share a Space that converts a lecture/paper to a quiz, and link in profile. Sponsorship route = HF blog/community spotlight, not paid ads.	ml-ai
3	r/GetStudying reddit.com/r/GetStudying/	~1.5M members	The largest study-method community — learners burned by passive re-reading/re-watching looking for active-recall systems, the exact pain Ritsu solves.	Reddit Ads (self-serve): run a Conversation/Image ad targeted by community (interest = r/GetStudying + lookalike study subs). Angle: 'Drop your textbook PDF → get an active-recall quiz in 30s' vs Anki/Quizlet. Reddit supports community-level interest targeting; pair with active-recall keyword targeting.	study-method

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
4	Art of Problem Solving (AoPS Community) artofproblemsolving.com/community	300,000+ members; 3.2M+ posts (used by most IMO winners as primary training resource)	The most rigor-obsessed, accuracy-demanding STEM learners (competition math, hard graded courses) — skeptics who will only adopt a tool that never lies to them.	No public ad manager — ORGANIC + partnership: seed in subject forums (Calculus, Olympiad), and pursue an AoPS-adjacent creator/instructor sponsorship. Lead with the accuracy/citations proof point (grounded in your own source) to win this skeptical crowd.	stem-course
5	r/learnmachinelearning reddit.com/r/learnmachinelearning/	~276K+ members (beginner/learner-focused ML sub)	Learners self-studying ML courses (Andrew Ng, fast.ai, cs231n) who explicitly want to MASTER, not cheat — the persona's core identity.	Reddit Ads: community-target r/learnmachinelearning + r/MachineLearning. Angle: 'Studying CS229/cs231n? Turn the lecture PDF into quizzes + flashcards + a concept map.' Use the course names verbatim in creative; A/B against NotebookLM/ChatGPT comparison.	ml-ai
6	DeepLearning.AI Community (Discourse) community.deeplearning.ai/	Large Discourse forum tied to Andrew Ng's courses (hundreds of thousands of DeepLearning.AI learners feed it; active AI Discussions + per-course categories)	Course-takers mid-way through dated, graded DeepLearning.AI specializations needing to convert long lectures into active practice before assessments.	Discourse, no native ads — ORGANIC: answer in per-course help categories and post a 'study companion' workflow for the specialization. Partnership angle = DeepLearning.AI 'The Batch' newsletter sponsorship to reach the same audience with a real ad unit.	ml-ai
7	fast.ai Course Forums forums.fast.ai/	Tens of thousands of practical-deep-learning students (long-running, high-engagement course forum)	Practitioners working lesson-by-lesson through fast.ai's multi-week course — precisely the deadline-paced, hands-on ML masterer.	Course forum, organic only — be helpful in lesson threads, share a notebook that turns the lesson video into a quiz. No paid unit; treat as credibility-seeding to fuel word-of-mouth into Discord/Reddit where paid scaling lives.	ml-ai
8	Learn AI Together (Discord) discord.com/invite/learnaitogether	~96K members	Largest beginner-friendly AI-learning Discord — active learners asking 'how do I actually learn this?', a warm audience for an active-mastery tool.	Discord has no platform-wide ad network — seed via server partnership/announcement (negotiate with admins) or run an AMA/study-tool demo in #resources. For paid scale, mirror the messaging into a Discord-ads-style creator sponsorship rather than a server CPC.	ml-ai
9	r/premed reddit.com/r/premed/	~200K+ members	Pre-meds in hard graded science courses (orgo, biochem, physics) with constant dated exams — high willingness-to-pay to replace passive study with active recall.	Reddit Ads: community-target r/premed. Angle: 'Orgo final in 3 days? Upload the chapter, get a quiz + concept map.' Lead with the exam-deadline timer; this audience already pays for UWorld/Anki, so position as a faster from-your-own-materials replacement.	exam-prep

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
10	r/MCAT reddit.com/r/MCAT/	~170K+ members	The most deadline-bearing, accuracy-obsessed exam-prep crowd; heavy Anki users actively seeking better active-recall tooling for a single high-stakes dated exam.	Reddit Ads: community-target r/MCAT. Angle: 'Turn your lecture notes / textbook into MCAT-style practice + flashcards.' Emphasize accuracy (no hallucinated questions) — this crowd will reject any tool that gets science wrong; that proof point is the whole ad.	exam-prep
11	r/learnmath reddit.com/r/learnmath/	~500K+ members (large math-learning help community)	Students grinding calculus, linear algebra and proof-based courses with graded problem sets — the STEM-course core of the wedge.	Reddit Ads: community-target r/learnmath + r/calculus + r/linearalgebra. Angle: 'Linear algebra midterm? Turn the lecture PDF into a quiz + concept map.' Use the specific course in creative; pair with 'active practice beats re-reading' learning-science hook.	stem-course
12	r/chemhelp reddit.com/r/chemhelp/	~100K+ members	Gen-chem and organic-chemistry students (a notorious weed-out course) facing dated exams — high pain, high substitution intent.	Reddit Ads: community-target r/chemhelp + r/OrganicChemistry. Angle: 'Orgo mechanisms not sticking? Upload the chapter → get drag-and-drop + quiz practice.' Highlight the diagram/mechanism activity types specific to chem.	stem-course
13	Cross Validated (Stats & ML Stack Exchange) stats.stackexchange.com/	Hundreds of thousands of users (largest Q&A site for statistics, ML, data analysis)	Rigor-demanding stats/ML learners verifying course concepts — accuracy-first skeptics who are the hardest-but-most-valuable to convert.	Stack Exchange offers limited display/sponsorship inventory, not granular self-serve community targeting — pursue a Stack Overflow/Stack Exchange network sponsored placement or, more cheaply, organic accurate answers that build credibility. Position on rigor + citations.	ml-ai
14	Physics Forums physicsforums.com/	Large established forum (operating since 2001; 'largest online physics/science community', grade-school through graduate level)	Physics and engineering students using the homework-help and coursework sections for dated graded problem sets — squarely on-persona for STEM courses.	Independent vBulletin/XenForo-style forum — direct banner/sponsorship deal with the forum owner is possible (no programmatic targeting). Cheaper path: organic helpfulness in homework-help subforums + signature link. Lead with 'active practice from your own lecture PDF'.	stem-course
15	r/GradSchool reddit.com/r/GradSchool/	~1M+ members	Grad students breaking down dense research papers and prepping for qualifying exams — the 'researcher/qualifying-exam' archetype of the wedge.	Reddit Ads: community-target r/GradSchool + r/AskAcademia. Angle: 'Qualifying exams? Turn a stack of papers into testable practice.' Emphasize the research-paper → quiz workflow and cross-session memory for long study horizons.	general-learning
16	Stack Overflow (+ Collectives/Discussions) stackoverflow.com/	Tens of millions of registered developers (largest programming Q&A community)	Engineers learning a new framework/language from docs under project deadlines — the 'engineer learning from docs, wants code exercises' archetype.	Stack Overflow Ads / Collectives sponsorship offers developer-audience targeting by tag/topic (e.g. python, pytorch, machine-learning). Angle: 'Learning PyTorch from the docs? Generate code exercises + a concept map from any doc page.' Tag-targeted display is the real paid unit here.	stem-course

#	Entity	Audience	Why this persona is here	How to target (ad / seed)	Cat
17	edX / MITx & Coursera course Discussion Forums edx.org/ (in-course discussion boards; e.g. MIT 6.x, Coursera ML)	Millions of enrolled learners across MITx/Coursera STEM MOOCs (per-course cohort forums, dated assignment deadlines)	Learners enrolled in graded, dated MOOC sessions (MIT 6.S191, Andrew Ng's Coursera ML) who need to convert long video lectures into active practice before deadlines.	In-platform forums are closed (no ads, no scraping) — reach this cohort via Google Search ads on course-name + 'study notes/quiz' queries and via the course's external creator channels (YouTube/newsletter). Treat the forum as the intent signal, not the ad surface.	general-learning
18	Student Doctor Network (SDN) Forums forums.studentdoctor.net/	Large long-running pre-health forum (hundreds of thousands of pre-med/med/grad-health members)	Pre-health students in brutal graded science sequences with dated board/course exams — overlaps the highest-WTP exam-prep slice of the wedge.	Independent XenForo forum — negotiate a direct banner/sponsored-thread placement with SDN (no programmatic targeting). Angle: 'Biochem/orgo exam in 3 days → quiz from your own notes.' Pair with the accuracy guarantee, which this clinical audience demands.	exam-prep

S10 · Pains, Triggers & Objections

(Ritsu icp-summary + JTBD · 20 attrs · 12★)

#	Attribute	Core Question	Answer	Conf	Source
★1	Top pain #1	<i>What hurts the most?</i>	Re-reads dense PDFs/slides for hours but it doesn't stick — passes nothing back, fails the midterm despite "studying"	Inf · 64%	icp-summary (passive method)
★2	Top pain #2	<i>What is the next-worst pain?</i>	Building own active-recall materials (Anki, practice problems) from a 200-page reader is hours of grunt work before any learning	Obs · 82%	analytics (PDF 63%) + icp
3	Top pain #3	<i>What else genuinely frustrates them?</i>	No structured PATH through a hard multi-week course — doesn't know what to master in what order before the deadline	Hyp · 50%	locked wedge (path = NotebookLM-gap POD)
★4	Pain intensity	<i>Is this hair-on-fire or mild annoyance?</i>	Hair-on-fire the 1–2 weeks before a graded midterm/final/qual; mild between deadlines (the WTP on/off cycle)	Inf · 64%	locked wedge (deadline-bearing)
5	Pain frequency	<i>How often do they feel this pain?</i>	Recurring on the course calendar — midterm + final + weekly problem sets; cohort = weekly graded modules (durable, not one-shot)	Inf · 64%	locked wedge
★6	Trigger event	<i>What event makes them act today?</i>	"Midterm in 5 days and I'm 3 lectures behind on cs231n / 6.S191" — or first failed problem set; uploads reader + lecture video that night	Obs · 80%	analytics (PDF+YT, sustained) + "ace exam in 3 days"
★7	Urgency level	<i>How urgent is it?</i>	Very high at trigger — fixed external deadline + real grade/credential stakes; not browsable "someday" learning	Inf · 64%	locked wedge (real stakes)
★8	Objection: trust/accuracy	<i>Do they doubt it will be accurate?</i>	"If it hallucinates one wrong quiz answer, it teaches me the wrong thing before my exam — I can't verify 50 cards" — the #1 kill risk	Inf · 66%	icp-summary + SERVQUAL

#	Attribute	Core Question	Answer	Conf	Source
★9	Objection: price	<i>Do they think it's worth the money?</i>	"\$29/mo when NotebookLM is free and ChatGPT Plus is \$20 / Quizlet \$3 / Anki \$0" — must justify a premium over a free clone of the spine	Inf · 64%	NotebookLM scan; deep-research stack prices
10	Objection: time	<i>Do they fear it takes too long to adopt?</i>	"I'm already behind — I don't have an afternoon to learn 17 activity types; I need value in the first session"	Hyp · 48%	JTBD (magic-moment <60s is the counter)
11	Objection: skepticism	<i>Do they dismiss it as hype?</i>	"Crosswords and emotional-AI personalities sound like edu-tainment fluff, not real mastery" — fears style over rigor	Inf · 60%	icp-summary (respects rigor)
12	Switching cost	<i>What does switching cost them?</i>	Low for net-new material; but already has Anki decks / Notion notes per course — re-uploading feels like redundant setup	Obs · 78%	analytics (32 projects, re-upload pattern)
★13	Status-quo bias	<i>Why is staying put easier?</i>	"Re-reading + highlighting got me through high school" — the failed passive method feels safe; mastery-tracking unproven to them	Inf · 60%	icp-summary (the FALSE belief)
★14	Cost of inaction	<i>What happens if they do nothing?</i>	A failed midterm/qual or a wasted \$2k bootcamp seat — real money + GPA/credential + the "I'm not smart enough" narrative hardens	Inf · 64%	locked wedge (real stakes)
★15	Deal-breakers	<i>What would instantly disqualify us?</i>	(a) any visible hallucinated quiz vs the source; (b) generation that breaks on dense PDFs/long lecture video; (c) no per-course mastery memory across sessions	Inf · 64%	SERVQUAL + analytics (PDF/long-video core)
★16	Must-haves	<i>What must be true for them to say yes?</i>	Source-grounded accurate quizzes; a structured multi-week path; cross-session mastery-tracking; flawless dense-PDF + long-YouTube ingest	Obs · 80%	analytics + locked wedge PODs
17	Objection: freemium-trap	<i>Do they suspect "free" is a bait-and-switch?</i>	"Is free real, or a paywall before any value?" — burned before; Free must deliver a full mastery loop on one real chapter before any wall	Inf · 56%	charter no-dark-patterns (VI-added)
★18	Evidence threshold	<i>How much proof before they trust it for a real exam?</i>	~2–3 accurate, source-cited quizzes on their OWN material before trusting it for a graded exam; one visible hallucination resets the counter	Hyp · 48%	SERVQUAL; icp (VI-added)
19	IP / copyright upload worry	<i>Are they nervous uploading their material?</i>	Real — they upload copyrighted textbooks + paid-bootcamp/proprietary slides; need an explicit "your material stays private, never trains public models" guarantee	Obs · 76%	analytics (pptx+PDF) + transparency.md (VI-added)
★20	Why-never-pay objection	<i>What makes them stay free forever?</i>	"NotebookLM + ChatGPT-free + Anki already cover me" — the soft-WTP reach layer rationalizes staying free; only a live graded deadline + a hard limit mid-path breaks it (this IS R1)	Hyp · 44%	R1 (WTP unproven) (VI-added)

S11 · Value, Pricing & WTP

(Kotler 4P-Price + customer-perceived-value + Ritsu substitution stack · 19 attrs · 9★)

#	Attribute	Core Question	Answer	Conf	Source
★1	WTP anchor	<i>What's the most they'd pay monthly?</i>	~\$15–25/mo ceiling for the committed masterer; Ritsu Plus \$29 sits just above anchor, so the deadline-window spike must justify the gap	Inf · 58%	icp-summary US anchor; product.md §10

#	Attribute	Core Question	Answer	Conf	Source
★2	Price sensitivity	<i>How much does price move their decision?</i>	High but deadline-elastic: normally resists, but a graded midterm in 7 days collapses sensitivity to near-zero for that window	Hyp · 50%	JTBD "exam in 3 days" timer
★3	Value drivers	<i>What specifically justifies paying?</i>	Time-to-mastery on dense PDF+lecture, accuracy (no hallucinated quizzes), the multi-week PATH, and cross-session progress they trust	Inf · 66%	analytics; SERVQUAL #1
4	Perceived value	<i>Does it feel worth it to them?</i>	"It turns my 600-page reader into a path I can actually finish" — values the active-mastery loop over passive tools	Hyp · 48%	icp-summary; "AI can't learn for you"
★5	Reference price	<i>What do they compare our price to?</i>	Anchors vs ChatGPT Plus / NotebookLM Pro ~\$20 and Google AI Pro student \$9.99/mo — that \$9.99 SKU is the hostile anchor	Inf · 64%	deep-research: NotebookLM Pro \$19.99, Google AI Pro student \$9.99
6	Substitution stack	<i>What are they paying for today, in total?</i>	ChatGPT Plus \$20 + Quizlet Plus \$7.99 + occasional Chegg \$14.95 ≈ \$25–43/mo; many run all-free (NotebookLM \$0 + ChatGPT free + Anki)	Inf · 62%	deep-research 2026 prices
7	Budget owner	<i>Who decides the spend?</i>	The learner personally (self-funded student/bootcamper); a minority parents/ employer-reimbursement, but the decision is self-driven	Hyp · 50%	icp-summary
8	Payment method	<i>How would they actually pay?</i>	Personal debit/credit or PayPal; student-tier expectations (Spotify, Google AI Pro \$9.99) frame "fair student price"	Hyp · 50%	deep-research student-SKU norms
9	Billing preference	<i>What billing shape do they prefer?</i>	Monthly first — buys for the course window, wary of annual lock-in pre-trust; annual –17% converts only after one full course delivers value	Inf · 60%	product.md annual –17%; episodic need
10	Discount sensitivity	<i>Does a discount change their decision?</i>	Responsive to a student/.edu discount or first-month-cheap, NOT manufactured urgency; \$9.99 Google AI Pro student is the bar	Inf · 60%	deep-research; charter no-dark-patterns
★11	Free-tier behavior	<i>How do they behave on free?</i>	Heavy free use until a real limit bites: multiple dense sources, many sessions, revisits (42 sources/36 sessions)	Obs · 82%	analytics live.sources/sessions
★12	Upgrade trigger	<i>What pushes them to pay?</i>	Hits the Free 40-page / 30-min-video / 5-session wall mid-course on a real graded deadline — the unproven R1 moment N=10 must test	Hyp · 42%	product.md Free limits; R1
★13	Cancel trigger	<i>What makes them stop paying?</i>	Course ends and no next deadline (cram-churn), OR a hallucinated quiz breaks trust, OR NotebookLM-free covers their need	Inf · 60%	R1 (cram-churn) + R2
14	Price-quality inference	<i>Does a higher price signal better to them?</i>	\$29 Plus reads "serious tool, not a toy" vs free clones — but ONLY if the <60s magic moment + accuracy deliver on first dense upload	Hyp · 48%	product.md "more for more"
15	ROI framing	<i>How do they rationalize the cost?</i>	"\$29 vs a failed midterm / a \$40–80/hr tutor / re-taking the course" — frames against grade-stakes + tutoring cost, not other apps	Inf · 62%	product.md replaces tutoring
16	Willingness to refer for value	<i>Would great value make them refer?</i>	High intent inside cohorts/Discords IF mastery lands; mechanism works but 100% founder-seeded — organic stranger referral unproven	Hyp · 46%	analytics shares=15 all founder
★17	Value-justification moment	<i>At what moment do they decide it's worth it?</i>	The instant a practice quiz on their own dense PDF exposes a real gap they'd re-read past — "it caught something I missed" — that's when \$29 stops feeling expensive	Hyp · 46%	JTBD; SERVQUAL (VI-added)

#	Attribute	Core Question	Answer	Conf	Source
18	Spend seasonality	<i>Do they only pay around exam season?</i>	Likely pays around graded windows (midterm/final/qual), at churn-risk between — the recurring-deadline cadence is what could make it durable vs one-shot cram	Inf · 56%	locked wedge (recurring) (VI-added)
★19	Pricing-pull-test status	<i>Is the \$29 WTP verified or still hypothesis?</i>	HYPOTHESIS — unverified at true-zero (0 real paying); SOP-PRODUCT-010 pricing-pull-test + the N=10 watch must verify the number before trusting it (this IS R1)	Hyp · 40%	R1; SOP-PRODUCT-010 (VI-added)

S12 · Competition & Substitution

(Kotler competitive frame + Ritsu · 12 attrs · 4★)

#	Attribute	Core Question	Answer	Conf	Source
★1	Current solution	<i>What's their current solution?</i>	Self-assembled stack: re-reading PDFs/slides + rewatching lecture YouTube + Anki + ChatGPT paste — passive, manual, no path	Inf · 64%	icp-summary; analytics (PDF 63% +YT 16%)
★2	Direct competitors	<i>Which direct rivals do they weigh?</i>	Google NotebookLM (free; doc→quiz/flashcard/Learning-Guide/track-mastery, Apr-2026) ; Quizlet; RemNote; smaller doc-to-quiz AI study apps	Inf · 66%	NotebookLM scan; deep-research
3	Indirect substitutes	<i>What non-obvious alternatives exist?</i>	Do-nothing (cram week-of); free ChatGPT/Gemini paste; the course's own problem sets + TA office hours + paid 1:1 tutoring	Hyp · 52%	JTBD; substitution stack
★4	Why current fails	<i>Why doesn't their current fix fully work?</i>	Passive re-reading feels productive but doesn't stick; ChatGPT/NotebookLM give answers/cards, not a graded multi-week mastery PATH with tracking	Inf · 66%	icp-summary; NotebookLM scan
5	Switching trigger	<i>What would make them switch to us?</i>	First failed/near-failed graded checkpoint on a hard ML/STEM course — proof that re-reading isn't working	Hyp · 50%	JTBD; deadline-bearing def
★6	Competitive dissatisfaction	<i>What annoys them about what they use?</i>	NotebookLM/ChatGPT are flat Q&A — no difficulty-ordered path, no 17 activity types, no 7 modes, no concept-map mastery progression over weeks	Inf · 64%	NotebookLM scan; feature delta
7	Brand awareness of Ritsu	<i>Do they even know we exist?</i>	Near-zero — true-zero launch, 0 organic stranger shares (all 15 founder-only); no Reddit/YouTube/#studytwt footprint yet	Obs · 86%	analytics live.shares
8	Brand perception	<i>What do they think of us today?</i>	Unformed; risk = bucketed as "another AI study gimmick" unless accuracy + learning-science lead the pitch	Hyp · 48%	icp-summary; SERVQUAL
9	Comparison criteria	<i>What do they compare tools on?</i>	Accuracy (no hallucinated quizzes) #1; then depth/path structure, activity variety, mastery-tracking, free-tier ceiling vs NotebookLM-free	Inf · 64%	SERVQUAL #1; NotebookLM-free benchmark
10	Displacement target	<i>Whose spend are we replacing?</i>	The \$25–100/mo study stack: Anki/Quizlet subs + ChatGPT Plus \$20 + occasional \$30–60/hr tutoring — consolidated into Plus \$29	Inf · 62%	product.md; deep-research
11	Share of wallet	<i>How much of their spend can we get?</i>	Targets near-100% of study-tool spend by replacing the stack; realistically first wins the ChatGPT-Plus \$20 + Quizlet slot	Hyp · 48%	JTBD; pricing

#	Attribute	Core Question	Answer	Conf	Source
★12	Category maturity	<i>Is the category new or familiar to them?</i>	Early-growth, fast-consolidating: NotebookLM (free, Sept-2025) commoditized the doc→quiz spine — table-stakes now, differentiation moving upstream to mastery-path	Inf · 66%	NotebookLM scan; deep-research

| S13 · Trust, Risk & Service Quality (SERVQUAL)

(Kotler SERVQUAL + product.md §6.7 · 14 attrs · 5★)

#	Attribute	Core Question	Answer	Conf	Source
★1	Reliability expectation	<i>How accurate must it be to trust it?</i>	Zero tolerance for hallucinated quiz/answer keys on graded cs231n/6.S191 material — a wrong key is worse than no quiz before a midterm	Inf · 66%	icp-summary + §6.7
★2	Error tolerance	<i>How forgiving are they of mistakes?</i>	Very low but recoverable: 1 bad item forgivable IF flagged + source-cited; 2–3 silent errors = uninstall, never paid	Hyp · 48%	Kotler SERVQUAL + JTBD
3	Responsiveness expectation	<i>How fast must it be?</i>	<30s to first activity from a 60-page PDF or long lecture video is table-stakes; dense PDF (63%) sets the latency bar	Obs · 80%	analytics live.sources
★4	Assurance needs	<i>What gives them confidence it's legit?</i>	Every explanation must cite the exact slide/page/timestamp from their own upload — "grounded, not guessed" vs ChatGPT	Inf · 64%	§6.7 + NotebookLM scan
5	Empathy needs	<i>Do they want it to feel encouraging?</i>	Counter "I'm not a math-science person" with an encouraging-but-rigorous tone (Supportive/Nerdy), not cutesy gimmicks	Inf · 60%	icp-summary
6	Tangibles	<i>Does the look-and-feel matter to them?</i>	Clean fast UI matters least of the five, but the <60s magic moment + heatmap must feel "serious tool," not consumer-cute	Hyp · 48%	§6.7 (tangibles lowest)
★7	Trust signals	<i>What proof makes them trust us?</i>	Visible source-grounding (page/timestamp citations), accuracy-first messaging, "AI can't learn for you" framing	Inf · 64%	positioning belief
8	Social proof type	<i>Whose endorsement moves them?</i>	Expert > peer > numbers: study-science creators (Sung, Abdaal, Frank) carry more weight than testimonials with a method-distrusting audience	Inf · 62%	channels; deep-research
9	Privacy concern	<i>Do they worry about data privacy?</i>	Moderate: cohort/bootcamp learners wary of accounts tied to graded coursework; want assurance data isn't sold or used to train public models	Hyp · 48%	JTBD + GDPR/CCPA
10	Data-safety concern	<i>Are they nervous uploading their material?</i>	Real: uploads copyrighted textbooks + paid-bootcamp/proprietary slides (pptx in mix) — need a clear "your material stays private" guarantee	Obs · 76%	analytics (pptx+PDF) + transparency.md
★11	Risk reversal needed	<i>What removes the risk of trying?</i>	Free \$0 must deliver a full mastery loop on one real chapter pre-paywall; the N=10 watch tests pay-at-first-limit	Inf · 62%	pricing + R1
12	Authority signals	<i>What authority cue earns belief?</i>	Name the method — active recall, spaced repetition, Feynman, concept maps — the preemptive moat NotebookLM lacks	Inf · 64%	positioning (learning-science)
13	Minor-safety (under-18)	<i>If some are under 18, what age-safety applies?</i>	A real edge: some HS-AP / younger learners exist; adult-default + age-gate + COPPA/age-handling (medical students are adults; high-schoolers are not)	Inf · 54%	product.md §12 (VI-added)

#	Attribute	Core Question	Answer	Conf	Source
★14	Error-recovery expectation	<i>When it gets one wrong, how must it handle it?</i>	They want a wrong item FLAGGED + source-cited + correctable, not silently confident — a graceful "here's the source, verify" beats false certainty	Hyp · 46%	SERVQUAL (VI-added)

| S14 · Customer Equity & Lifecycle

(Kotler CLV/equity + 5-Star Module 11 · 15 attrs · 8★)

#	Attribute	Core Question	Answer	Conf	Source
1	Expected CLV	<i>What is one customer worth over time?</i>	~\$200–350 over a 7–10mo lifespan: Plus \$29/mo retained ~5–7 academic-cycle months, minus ~10% fees; pre-PMF, unproven	Hyp · 40%	Kotler CLV + pricing; WTP unproven
★2	Retention likelihood	<i>How likely are they to retain?</i>	Moderate-to-high WHILE a graded deadline is live (recurring midterms re-trigger value); drops sharply between terms	Inf · 58%	icp-summary (recurring deadline)
★3	Churn risk factors	<i>What signals they'll churn?</i>	Course ends/term break; one hard limit before value proven; NotebookLM-free "good enough"; a hallucinated quiz breaks accuracy trust	Inf · 62%	NotebookLM scan + SERVQUAL + R2
4	Expansion potential	<i>Can they grow into a higher tier?</i>	Real: dense ML PDFs + 6h lecture videos blow Free's 40pg/30min + Plus's 100pg/2h, pulling Plus→Pro (\$59, unlimited, export)	Obs · 80%	analytics (PDF 63% +YT 16%) + tiers
★5	Referral propensity	<i>How likely are they to refer?</i>	Mechanism works (15 shares) but ZERO organic stranger referral yet — all founder; cohort context favors latent virality	Obs · 86%	analytics live.shares
6	Advocacy archetype	<i>What kind of champion would they become?</i>	"The cohort study-buddy evangelist" — shares the Knowledge-Map link inside their class Discord; proof-by-grade, not hype	Hyp · 46%	Kotler 5A + cohort JTBD
7	NPS expectation	<i>Would they score us a promoter?</i>	High among masterers who hit the aha (heavy-revisit users) — target ≥40 for PMF; suppressed by accuracy slips	Inf · 56%	icp-summary PMF gate
★8	Loyalty driver	<i>What would keep them with us?</i>	Cross-session mastery-tracking + structured multi-week PATH (the NotebookLM-proof moat) — switching means losing accumulated progress	Inf · 64%	locked moat
★9	Engagement depth	<i>How deep does their usage go?</i>	Deep for the committed core: 42 sources/36 sessions, 4 sources/10 sessions (heavy revisit); 3329 units from uploads; small-N	Obs · 84%	analytics live.sessions
10	Habit-formation likelihood	<i>Will it become a habit for them?</i>	High during a term (daily/weekly graded cadence + heatmap + streaks form a ritual); fragile across the inter-term gap	Inf · 58%	Ritsu heatmap + deadline-cadence
★11	Lifecycle stage	<i>Where are they in the lifecycle?</i>	Acquisition/activation — true-zero, 0 real paying; the core is unvalidated on PAYMENT, only on engagement	Obs · 86%	analytics (0 real paying)
12	Reactivation potential	<i>Could we win them back if they leave?</i>	High + predictable: next term's hard course/qual is a natural re-entry; lapses are seasonal, not abandonment	Inf · 60%	recurring-deadline WTP
13	Win-back trigger	<i>What reactivation offer would work?</i>	"New semester, new dense course" email + a fresh syllabus-PDF→PATH demo; or "your Knowledge Map is waiting" cross-session pull	Hyp · 48%	Module 11 + cross-session memory

#	Attribute	Core Question	Answer	Conf	Source
★14	Equity quadrant	<i>Which value × loyalty quadrant are they in?</i>	TRUE-FRIENDS aspiration (high-fit, recurring need + loyalty); currently STRANGERS in reality (0 paid, engagement-only); N=10 must convert	Inf · 60%	Kotler equity matrix + true-zero
★15	Sean-Ellis fit	<i>Would ≥40% be "very disappointed" without Ritsu?</i>	The PMF gate — UNMEASURED at true-zero; the segment is <i>designed</i> to clear ≥40% (durable need + 1:1 fit) but it's the central PMF test (SOP-GTM-016)	Hyp · 44%	SOP-GTM-016 (VI-added)

S15 · Ritsu-Specific Product Fit (deep)

(Ritsu product.md + icp-summary · 26 attrs · 11★)

#	Attribute	Core Question	Answer	Conf	Source
1	Source-material type	<i>What raw material do they bring?</i>	Dense PDF course-packs/textbook chapters (63% of uploads) + long lecture videos (YouTube 16%) + lecture slides (pptx) + code	Obs · 86%	analytics live.sources
★2	Primary activity fit	<i>Which of 17+ activities is their hook?</i>	Quiz + Flashcard (spaced recall) anchor; Code-exercise for ML; Diagram/Mindmap/Concept-map for cross-chapter linking	Inf · 64%	17-activity set + STEM JTBD
3	Tutoring-mode fit	<i>Which of 7 modes fits them?</i>	"Ask Me" (explain-back = Feynman test of mastery) primary; "Exercise"/"Solve" for problem-sets; "Adaptive" across the path	Inf · 60%	icp + 7 modes
4	Subject / domain	<i>What subject do they bring?</i>	ML/AI courses (cs231n, 6.S191, Andrew Ng, fast.ai) = beachhead; broader STEM (calc/orgo/physics/bio, pre-prof) = expansion	Inf · 68%	locked wedge
5	Use-case workflow	<i>Which situational job is theirs?</i>	"Master a textbook chapter" + "Learn from a YouTube lecture" + "Learn a new framework fast" stitched across a graded multi-week course	Inf · 64%	homepage jobs + wedge
6	Mastery goal	<i>What does mastery look like for them?</i>	True understanding to pass graded midterm/final/qual — mastery non-optional because the deadline has real stakes	Inf · 64%	wedge def
★7	Magic-moment definition	<i>What is their precise first-win moment?</i>	Upload one cs231n lecture PDF → in <60s a structured path + a first quiz that exposes a real gap they re-read past	Inf · 64%	magic-moment <60s + icp
★8	Activation definition	<i>What counts as activated for them?</i>	Reaches first-quiz aha AND returns for a 2nd session on the same source within 7 days (revisit = mastery intent)	Obs · 80%	analytics heavy-revisit users
★9	Wedge-fit hypothesis	<i>Does the wedge actually pull this segment?</i>	1:1 fit — recurring graded deadlines = durable WTP trigger (vs one-shot cram churn); risk: PAYMENT unproven	Hyp · 44%	MASDA rationale + R1
10	Personality-mode pref	<i>Which tutor personality do they want?</i>	"Nerdy" + "Supportive" (rigor-respecting, anti-gimmick); "Funny/Sarcastic" repel this distrust-gimmicks learner	Hyp · 46%	icp + 10 styles
11	Device pattern	<i>Phone-first or laptop-deep?</i>	Laptop-deep for the multi-week mastery work; phone only for flashcard micro-review between classes	Inf · 62%	session depth + STEM workflow
12	Persona mapping	<i>Which of the 6 personas is this?</i>	Primary = Students (graded STEM); secondary = Self-Learners + Engineers (framework-from-docs); Researchers on the expansion edge	Inf · 64%	6 marketing personas + wedge

#	Attribute	Core Question	Answer	Conf	Source
13	Tier fit	<i>Which pricing tier fits them?</i>	Plus \$29/mo (most-popular; 100pg/2h-video/20-session fit a dense multi-week course); Pro \$59 only if cross-course sharing/export	Inf · 62%	pricing tiers + material density
14	Credit-usage pattern	<i>How heavily would they burn credits?</i>	Spiky-then-sustained: heavy burst at each new chapter/lecture upload, steady recall draws between — first hard limit hits mid-course at deadline	Hyp · 46%	credit model + 3329 units/756 sources
★15	Share-link behavior	<i>Would they share a Ritsu link?</i>	Shares a generated quiz/Knowledge-Map to study-group; mechanism works but unproven on strangers (15 shares, all founder)	Obs · 82%	analytics (15 shares, 0 organic)
★16	Counter-positioning	<i>Why Ritsu over their current stack?</i>	vs ChatGPT: persistent structured PATH not one-off chat; vs Anki/Quizlet: auto-built from source; vs NotebookLM: the multi-week path + 17 activities + cross-session mastery-tracking it LACKS	Inf · 64%	R2 NotebookLM scan + differentiators
★17	Quiz fit	<i>Is auto active-recall quizzing their core need?</i>	Core — auto active-recall quizzing from their own dense source is the primary hook (the #1 adopted activity); active recall is the named learning-science	Inf · 64%	product §6.2 (VI-added)
18	Flashcard fit	<i>Does auto-flashcard replace their manual Anki?</i>	Strong — auto-flashcards remove hours of manual Anki card-making (their #2 pain); spaced repetition is already familiar + trusted	Inf · 62%	icp stack (VI-added)
19	Mindmap / Knowledge-Map fit	<i>Do they value the concept-map of gaps?</i>	High for a multi-week course — the concept-map surfaces cross-chapter gaps (a NotebookLM-gap POD); Basic at Plus, Full (share+export) at Pro	Inf · 60%	product §6.4 (VI-added)
20	Code-exercise fit	<i>Do ML/engineers need code-exercise from docs?</i>	Relevant for the ML/engineer slice — code-exercise from framework docs (fast.ai, PyTorch) is a distinct hook generic study tools lack	Inf · 58%	product §6.2 (VI-added)
21	Diagram / Timeline fit	<i>Do diagram/timeline suit their material?</i>	Situational — diagram suits systems/architectures, timeline suits process/history; secondary to quiz/flashcard for math-heavy ML	Hyp · 44%	product §6.2 (VI-added)
★22	Ask-Me (Feynman) mode fit	<i>Does explain-back mastery-test them?</i>	High — "explain it back" IS the mastery test this persona respects; the Feynman framing signals the learning-science rigor they trust	Inf · 60%	product §6.3 (VI-added)
23	Adaptive mode fit	<i>Do they want difficulty that adapts to results?</i>	Strong — they want difficulty that adapts to where they're actually weak across the multi-week path, not fixed drills	Inf · 58%	product §6.3 (VI-added)
★24	Multi-week PATH fit	<i>Is the ordered path the core differentiator?</i>	THE load-bearing POD — the difficulty-ordered, tracked, multi-week PATH is exactly what free doc-Q&A tools (NotebookLM) lack; it carries the wedge	Inf · 66%	locked moat (VI-added)
★25	Per-source-limit fit	<i>Do dense sources blow the Free/Plus page-limit?</i>	Yes — 200-pg readers + 6h lecture series blow Free's 40pg/30min and pressure Plus's 100pg/2h → the natural Plus→Pro expansion + the upgrade trigger	Obs · 78%	analytics + pricing limits (VI-added)
★26	Counter-positioning vs NotebookLM	<i>Why Ritsu over free NotebookLM specifically?</i>	Ritsu wins ONLY on the structured multi-week PATH + 17 activity types + concept-level Knowledge Map + named learning-science — NEVER on "we also make quizzes" (NotebookLM does that free); R2 = this residual moat may not be enough	Inf · 60%	R2 NotebookLM scan (VI-added)

S16 · Measurement & Validation (MASDA)

(Kotler Ch7 MASDA + Ritsu observable proxies · 17 attrs · 10★)

#	Attribute	Core Question	Answer	Conf	Source
★1	Measurable	<i>Can we actually measure this segment?</i>	Sizeable + countable: enrolled US/English STEM-ML learners in a graded course — proxy via enrollments (cs231n ~600/yr, fast.ai, Coursera ML)	Inf · 62%	deep-research; icp-summary
★2	Accessible	<i>Can we reach them at reasonable cost?</i>	Pre-assembled on r/learnmachinelearning, r/MachineLearning, fast.ai forums, #studytwit, Sung/Abdaal audiences — reachable today, zero net-new channel build (S9-A = 262 mapped placements)	Inf · 64%	icp-summary channels; S9-A
★3	Substantial	<i>Is it large enough to be worth it?</i>	Worth pursuing: the 100-paying north-star needs only a thin slice; replaces a \$25–100/mo stack, so per-user economics clear \$29 easily	Inf · 62%	north-star + pricing
★4	Differentiable	<i>Do they behave distinctly enough to target?</i>	Distinct from exam-crammer (churns post-exam) and soft-WTP open self-learner: recurring graded deadlines = durable repeat-pay trigger, behaviorally separable	Inf · 62%	locked wedge + JTBD
★5	Actionable	<i>Can a solo + AI team actually serve them?</i>	One coherent offer fits 1:1 (doc→path→17 activities→mastery-tracking); single ICP message, single channel set	Inf · 64%	locked wedge + capability fit
★6	Observable proxy	<i>What signal predicts a high-value member?</i>	Heavy revisit on dense material: ≥10 sessions across ≥1 PDF/long-video over multiple weeks (top users: 42src/36sess and 4src/10sess)	Obs · 82%	analytics sources × sessions
★7	Predictive signal	<i>Which action predicts they'll pay?</i>	Hitting the Free hard limit (600 credits / 40pp / 30min / 5 sessions) while mid-course-path — the "blocked at the first hard limit" moment	Hyp · 44%	Free-tier limits + JTBD
8	Data source	<i>Where does the data come from?</i>	supabase-analytics pseudonymized <i>live.*</i> (sources, learning_sessions, learning_units, session_shares, payments, credit_transactions) on user_hash	Obs · 88%	analytics live schema (30 views)
9	Confidence level	<i>How sure are we — guess or observed?</i>	Engagement/material/mastery-pattern = observed at small-N (25 users); PAYMENT + stranger-referral = unproven (0 real paying, 15 founder shares)	Obs · 86%	analytics
★10	Validation method	<i>How will we validate this?</i>	N=10 stranger watch (SOP-PRODUCT-002) testing pay-at-first-limit + 1:1 interviews; later activation→paid telemetry + pricing-pull A/B (SOP-PRODUCT-010)	Inf · 62%	SOP-PRODUCT-002/010
★11	N observed	<i>How many real ones have we watched?</i>	25 pseudonymized profiles; ~24 ex-founder; only 2–3 sustained-use (42/36, 17/13, 4/10 sess); 0 paying strangers — true-zero, tiny-N	Obs · 88%	analytics live.profiles (25)
12	Last validated	<i>When did we last verify this?</i>	2026-06-04 (analytics sync v0.17.2, 30 synced views incl. payments); behavioral pattern re-queried live 2026-06-06	Obs · 88%	manifest + this session
★13	Falsifiability	<i>What would prove this segment false?</i>	N=10 don't pay at the first hard limit, OR NotebookLM's free doc→quiz spine satisfies the JTBD so deadline-bearers won't pay the Plus premium	Inf · 60%	R1 (WTP) + R2 (NotebookLM)
14	Stability	<i>How quickly does this segment change?</i>	Core JTBD (master-hard-course-by-deadline) is durable for years; the ML-beachhead surface drifts fast (courses rotate yearly) — re-pin named courses quarterly	Inf · 62%	JTBD + beachhead/expansion split
★15	KPI — aha < 60s	<i>Do they hit the <60s magic moment?</i>	Target ≥X% of new signups reach the first-quiz aha in <60s on their OWN dense material; 30s path generation works, but the stranger aha-RATE is unmeasured — the N=10 watch's 1st leading metric	Hyp · 46%	magic-moment + SOP-PRODUCT-002 (VI-added)
★16	KPI — free→paid > 5%	<i>Does free→paid clear 5%?</i>	Target >5% free→paid for this segment; observed 0% (0 real paying) — the single most important unproven number (R1)	Hyp · 40%	R1; north-star (VI-added)

#	Attribute	Core Question	Answer	Conf	Source
★17	Predictive pay-signal	<i>Which single action best predicts paying?</i>	Hitting the first hard credit/page limit WHILE mid-path with a live deadline (the "blocked mid-momentum" moment); the N=10 watch instruments exactly this	Hyp · 44%	R1; JTBD (VI-added)

S17 · Business / Firmographic (B2B educator expansion — DEFERRED, not first-100)

(Kotler Ch6 business markets — expansion only · 12 attrs · 0★ for the wedge)

Read this section as the EXPANSION ladder, not the wedge. The first-100 is pure B2C (buyer = user = payer). The B2B educator/institution path is a post-PMF land-and-expand that the B2C product seeds bottom-up (PLG). Every row here is honestly low-confidence — it is a *future* motion, not an observed one.

#	Attribute	Core Question	Answer	Conf	Source
1	Organization type	<i>What kind of org buys for them?</i>	Universities (depts/professors), ML/data bootcamps (fast.ai-style), corporate L&D, test-prep centers — the educator persona creating courses for students	Hyp · 42%	product.md §8 (Educators)
2	Org size	<i>How big is the buying organization?</i>	Wide — a single professor/TA (1 seat) → a department (10s) → a bootcamp cohort (100s); land tiny, expand by course	Hyp · 40%	Kotler Ch6
3	Industry / vertical	<i>What vertical is the org in?</i>	Higher-ed STEM/CS departments, ML/data bootcamps, corporate engineering upskilling	Hyp · 40%	beachhead → expansion
4	Buying-center roles	<i>Who plays each buying role?</i>	Initiator/user = instructor; influencers = students; decider/budget = dept head or L&D lead; gatekeeper = procurement/IT	Hyp · 38%	Kotler Ch6 buying center
5	Decision process	<i>How many steps to a signed deal?</i>	Long + multi-approval for institutions (Kotler Ch6) — BUT PLG bottom-up: a single instructor self-serves first, then advocates up	Hyp · 40%	Kotler Ch6 + PLG
6	Decision timeline	<i>How long is their buying cycle?</i>	Months for institutional procurement; a single educator can adopt in days (self-serve) then champion internally	Hyp · 40%	Kotler Ch6
7	Budget cycle	<i>When does their budget free up?</i>	Academic-year + semester budget cycles; corporate L&D annual	Hyp · 40%	Kotler Ch6
8	Procurement constraints	<i>What procurement hoops exist?</i>	Data-privacy / FERPA (student data), security review, accessibility (ADA), maybe SSO — real institutional hurdles	Inf · 48%	Kotler Ch6 + 07-trust-safety
9	Derived demand	<i>What downstream need pulls their purchase?</i>	Student outcomes (pass rates, engagement) + instructor time saved making materials	Hyp · 42%	product.md §8
10	Seat / license model	<i>How would they license it?</i>	Per-seat (student) or site license; a Business plan already exists on the pricing page	Inf · 48%	product.md §10 (Business plan)
11	Champion profile	<i>Who internally would champion us?</i>	An enthusiastic instructor whose students loved the B2C product — the bottom-up PLG champion	Hyp · 42%	PLG motion

#	Attribute	Core Question	Answer	Conf	Source
12	Expansion path	<i>How does a first deal expand?</i>	One instructor → their course → the department → the institution (land-and-expand); the B2C wedge seeds the B2B champion	Hyp · 44%	PLG land-and-expand

Confidence ledger + what to do with this

- **observed (the spine of truth at true-zero)**: material mix (PDF 63% + YouTube 16%), the sustained-revisit mastery pattern (42 src/36 sess), 3,329 generated units, 15 founder-only shares (⇒ 0 organic stranger-referral), 0 real paying. Everything else is **inferred** (market deep-research / Kotler+BRAVO frameworks) or **hypothesis** (untested). The **Conf %** on every row is an honest P(answer correct): observed ≈ 80–92%, inferred ≈ 55–72%, hypothesis ≈ 30–50%.
- **The two unproven load-bearing claims** the whole persona rests on: **(R1)** that deadline-bearing masterers *pay* at the first hard limit (the N=10 watch's central test — S16 §16/§17, S11 §19, S10 §20 all sit at Conf ≤ 44%), and **(R2)** that Ritsu's PATH + mastery-tracking + learning-science out-hold *free* NotebookLM (which **narrowed the gap further in April 2026** — adding mastery-tracking + a Socratic "Learning Guide"; Ritsu's honest residual moat = the **structured multi-week ordered PATH + 17 distinct activity types + concept-level Knowledge Map + named learning-science** — S15 §24/§26).
- **What changed in v2 (this revision)**: added the **Core Question + Confidence(%)** columns to every attribute; expanded to all **17 sections / ~307 attributes** (union of the EN-266 base + the Ritsu-tuned VI-308 template — adding S17 B2B + ~50 Ritsu-specific questions the prior draft lacked); replaced the thin S9 with **S9-A**, a 262-entity, ad-targetable channel & location map (11 platforms, ~30 each for YouTube/Reddit/X/Facebook/Instagram) built by a parallel research workflow — paste it straight into ad managers.
- **How to use S9-A for ads**: the 5 core platforms (YouTube/Reddit/X/Facebook/Instagram) carry ~30 specific placements each, with the exact ad mechanism per row (YouTube managed-placement, Reddit community targeting, X follower-lookalike, Meta interest/group, etc.). Discord/forums = the organic Collison-install recruiting map for the N=10 (SOP-PRODUCT-002), not paid placements. Authority > raw reach for this skeptical audience — lead with learning-science creators (Justin Sung, Ali Abdaal, StatQuest), not mass-blasts.
- **N=10 recruiting profile (for SOP-PRODUCT-002)**: recruit **deadline-bearing committed masterers** — students/switchers in a hard, graded, multi-week ML/STEM course (cs231n / 6.S191 / Andrew Ng / fast.ai / a bootcamp cohort / grad quals) with a *dated* graded deadline; from r/learnmachinelearning, r/MachineLearning, fast.ai forums, course Discords, #studytwit (see S9-A → Reddit/Discord/Forums). Watch the one thing that matters: **do they pay at the first hard limit?**

*Sources consulted (external): Coursera 2025 Global Skills Report; Stanford cs231n / CS229; fast.ai; Andrew Ng courses; r/MachineLearning + r/learnmachinelearning member counts; NotebookLM (blog.google flashcards/quizzes + student features + April-2026 update); ChatGPT/Quizlet/Chegg/Google-AI-Pro 2026 pricing; the S9-A channel-research workflow (11 platforms, 262 entities, web-verified 2026-06-06). Internal: supabase-analytics Door-2 live.; product.md; positioning.md; icp-summary.md; north-star.md; brand_voice.md.**